

PART I

2026-06-05

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##### Part I - Cross-Sectional OLS #####
##### Insurance and U.S. State Mortality #####
#####
#####
###
### Research question: across U.S. states, is mortality associated with
### health-insurance coverage, once age structure and economic conditions
### are accounted for?
###
### Design: observational, cross-sectional, aggregate (state-level) data
### for 2019 (n = 51: 50 states + D.C.). Coverage is NOT randomly assigned,
### so the estimates are associations, not causal effects.
###
### Main method: progressive OLS model building (mdl_1 -> mdl_5) used as an
### omitted-variable-bias diagnostic; HC3 heteroskedasticity-robust
### inference plus joint F-tests on the insurance and income terms.
###
### Outcome: starts from crude_rate_mortality (Models 1-4) and converges on
### age_adjusted_mortality as the PREFERRED outcome (Model 5).
###
### Data: 2019 U.S. state-level data (n = 51), assembled in the data-prep
### step; analysis input: mio_dataset_output.csv.
###
### Note on sources: commands taught in the course are used natively;
### methods beyond the Labs (e.g. HC3 robust SE) are flagged inline with
### their source.
###
#####

# ---- Packages -----
library(car)          # LinearHypothesis (joint F-tests)

## Loading required package: carData

library(sandwich)     # vcovHC (HC3 robust SE)
library(lmtest)       # coeftest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric
```

```
library(lattice)      # levelplot (correlation heatmaps)
library(viridisLite) # viridis palette (heatmaps)

dt <- read.csv("mio_dataset_output.csv", sep = ",", dec = ".", header = TRUE)
print(head(dt, 10))
```

```
##      GEOID      NAME median_income median_age poverty_per_100k
## 1      1      Alabama      51734      39.4      15500
## 2      2      Alaska      75463      35.0      10100
## 3      4      Arizona      62055      38.3      13500
## 4      5      Arkansas      48952      38.8      16200
## 5      6      California      80440      37.0      11800
## 6      8      Colorado      77127      37.1      9300
## 7      9      Connecticut      78833      41.2      10000
## 8     10      Delaware      70176      41.4      11300
## 9     11 District of Columbia      92266      34.3      13500
## 10    12      Florida      59227      42.4      12700
##      insurance_per_100k crude_rate_mortality age_adjusted_mortality
## 1      90300      1103.5      897.8
## 2      87800      630.6      700.9
## 3      88700      827.6      660.5
## 4      90900     1089.8      876.1
## 5      92300      682.9      601.8
## 6      92000      684.0      647.8
## 7      94100      890.4      648.7
## 8      93400      955.3      728.6
## 9      96500      698.1      703.2
## 10     86800      963.8      649.4
##      crude_rate_medical_mortality age_adjusted_medical_mortality
## 1      734.3      591.5
## 2      407.2      449.4
## 3      568.7      450.2
## 4      741.5      589.9
## 5      502.0      440.2
## 6      454.2      428.0
## 7      577.3      420.3
## 8      663.2      502.8
## 9      482.6      488.3
## 10     667.7      446.0
```

```
summary(dt)
```

```
##      GEOID      NAME median_income median_age
## Min.   : 1.00   Length   :51   Min.    :45792   Min.    :31.20
## 1st Qu.:16.50   N.unique :51   1st Qu.:57506   1st Qu.:37.70
## Median :29.00   N.blank  : 0   Median :63229   Median :38.80
## Mean   :28.96   Min.nchar: 4   Mean   :65511   Mean    :38.79
## 3rd Qu.:41.50   Max.nchar:20   3rd Qu.:75028   3rd Qu.:39.85
## Max.    :56.00           Max.    :92266   Max.    :45.10
## poverty_per_100k insurance_per_100k crude_rate_mortality
## Min.    : 7300   Min.    :81600   Min.    : 584.4
## 1st Qu.:10100   1st Qu.:89850   1st Qu.: 825.8
```

```
## Median :11800      Median :92000      Median : 930.5
## Mean   :12171      Mean    :91600      Mean    : 911.0
## 3rd Qu.:13500      3rd Qu.:93850      3rd Qu.: 989.0
## Max.   :19600      Max.    :97000      Max.    :1305.9
## age_adjusted_mortality crude_rate_medical_mortality
## Min.   :573.3      Min.    :372.3
## 1st Qu.:692.1      1st Qu.:547.6
## Median :721.6      Median :605.4
## Mean   :741.5      Mean    :609.7
## 3rd Qu.:772.0      3rd Qu.:673.8
## Max.   :945.4      Max.    :867.6
## age_adjusted_medical_mortality
## Min.   :386.6
## 1st Qu.:453.8
## Median :483.5
## Mean   :493.1
## 3rd Qu.:517.1
## Max.   :643.8
```

*# Rescale: insurance and poverty are already per-100,000; convert to rates (0-100).
So coefficients are interpretable as "per 1 pp change in insured share".*

```
dt$insured_rate <- dt$insurance_per_100k / 1000
dt$poverty_rate <- dt$poverty_per_100k / 1000
dt$log_income <- log(dt$median_income)
dt$age <- dt$median_age
dt$mortality <- dt$crude_rate_mortality
dt$agemortality <- dt$age_adjusted_mortality
dt$medical_mortality <- dt$crude_rate_medical_mortality
dt$age_adjusted_medical_mortality_rate <- dt$age_adjusted_medical_mortality
```

MEAN-CENTERING FOR INTERPRETABLE INTERACTIONS

*# Centering variables around their sample means allows main effect coefficients
to retain vital structural economic meaning in the presence of interaction terms.*

```
dt$c_insured_rate <- dt$insured_rate - mean(dt$insured_rate)
dt$c_poverty_rate <- dt$poverty_rate - mean(dt$poverty_rate)
dt$c_log_income <- dt$log_income - mean(dt$log_income)
dt$c_age <- dt$age - mean(dt$age)
```

```
print(summary(dt[, c("mortality", "insured_rate", "poverty_rate",
                    "age", "log_income")]))
```

```
## mortality      insured_rate      poverty_rate      age
## Min.   : 584.4      Min.    :81.60      Min.    : 7.30      Min.    :31.20
## 1st Qu.: 825.8      1st Qu.:89.85      1st Qu.:10.10      1st Qu.:37.70
## Median : 930.5      Median :92.00      Median :11.80      Median :38.80
## Mean    : 911.0      Mean    :91.60      Mean    :12.17      Mean    :38.79
## 3rd Qu.: 989.0      3rd Qu.:93.85      3rd Qu.:13.50      3rd Qu.:39.85
## Max.    :1305.9      Max.    :97.00      Max.    :19.60      Max.    :45.10
## log_income
## Min.    :10.73
## 1st Qu.:10.96
```

```
## Median :11.05
## Mean   :11.08
## 3rd Qu.:11.23
## Max.   :11.43
```

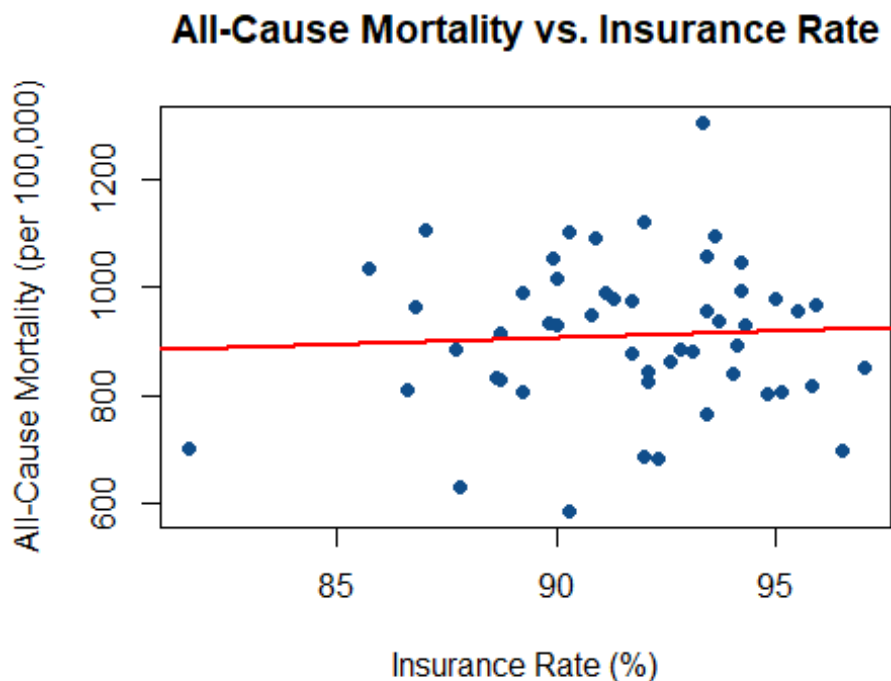
```
print(round(cor(dt[, c("mortality", "insured_rate", "poverty_rate",
                      "age", "log_income")],
              use = "complete.obs"), 3))
```

```
##          mortality insured_rate poverty_rate    age log_income
## mortality      1.000      0.059      0.517  0.641    -0.734
## insured_rate    0.059      1.000     -0.379  0.316     0.447
## poverty_rate    0.517     -0.379      1.000 -0.019    -0.782
## age            0.641      0.316     -0.019  1.000    -0.176
## log_income     -0.734      0.447     -0.782 -0.176     1.000
```

We want to study whether mortality (first crude, then adjusted) is affected by insurance and other control variables.

We start by plotting the main variables of interest.

```
plot(dt$insured_rate, dt$mortality,
     pch = 16, col = "dodgerblue4",
     main = "All-Cause Mortality vs. Insurance Rate",
     xlab = "Insurance Rate (%)",
     ylab = "All-Cause Mortality (per 100,000)")
abline(lm(mortality ~ insured_rate, data = dt), col = "red", lwd = 2)
```



```

# Note that there are more observations at high levels of the insurance rate; this is not a measurement error.
median(dt$insured_rate)

## [1] 92

# This means that citizens in every state are, in the majority, covered by some type of insurance.
# Taking only one state per rate would lead to overestimation for the less-insured states,
# with the finding being driven by that.

# Then we proceed by inspecting, at first sight, the possible correlations between the variables of the dataset.
# From this matrix we can see if our assumptions are plausible for a regression/multi-stage regression.

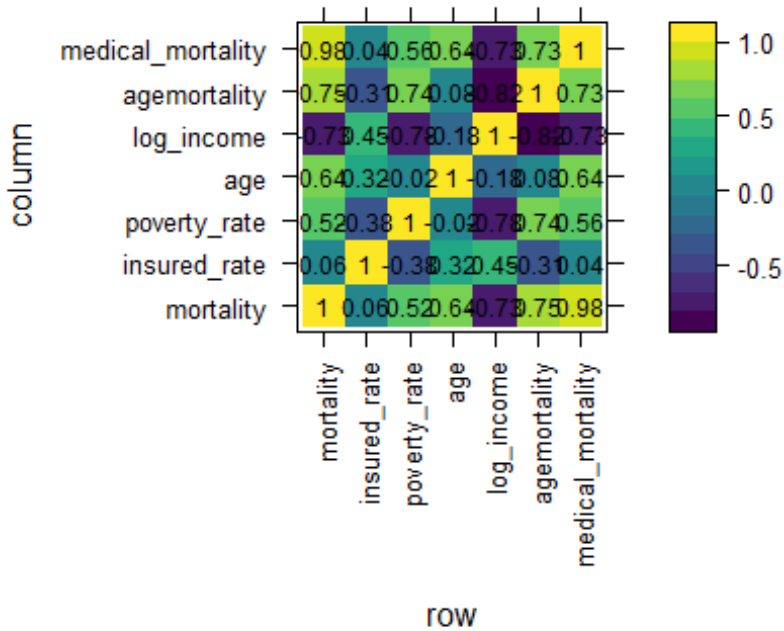
cor_mat <- cor(dt[, c("mortality", "insured_rate", "poverty_rate", "age", "log_income", "agemortality", "medical_mortality")])

# Panel function
myPanel <- function(x, y, z, ...) {
  lattice::panel.levelplot(x, y, z, ...)
  my_text <- ifelse(!is.na(z), paste0(round(z, 2)), "")
  lattice::panel.text(x, y, my_text, cex = 0.75)
}

# Full matrix
lattice::levelplot(cor_mat,
                  panel = myPanel,
                  col.regions = viridisLite::viridis(100),
                  main = "Correlation matrix - all variables",
                  scales = list(x = list(rot = 90)))

```

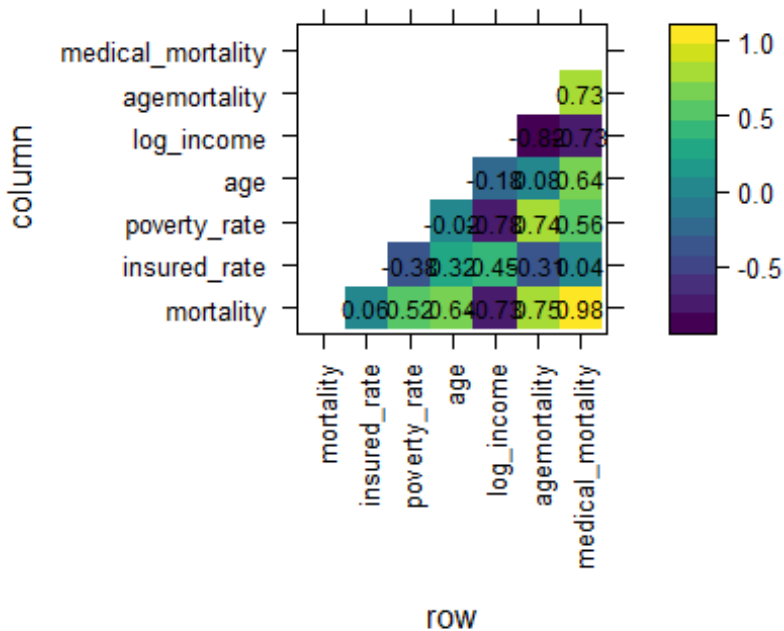
Correlation matrix – all variables



```
# Lower triangle only
mask <- cor_mat
mask[upper.tri(mask, diag = TRUE)] <- NA

lattice::levelplot(mask,
  panel = myPanel,
  col.regions = viridisLite::viridis(100),
  main = "Correlation matrix (lower triangle)",
  scales = list(x = list(rot = 90)))
```

Correlation matrix (lower triangle)



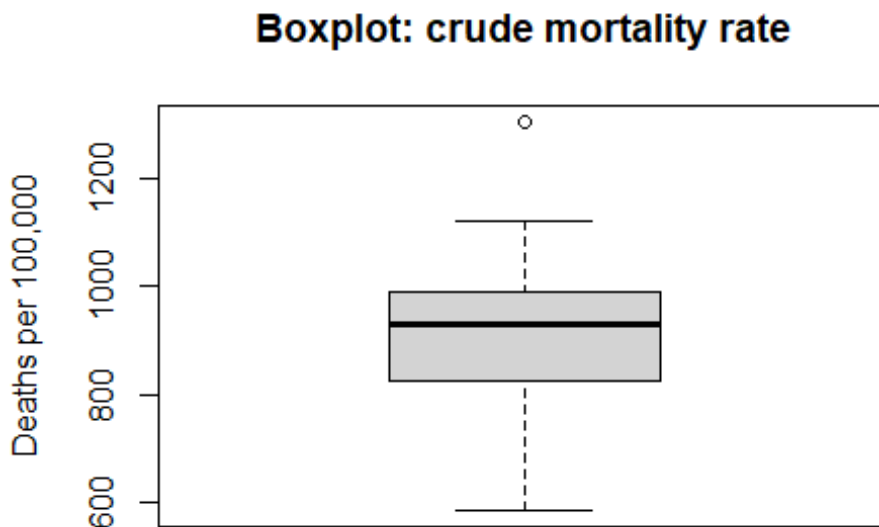
```
# BIVARIATE BASELINE (Model 1)
# By simple reasoning, we expect insurance to have a positive effect on health and
# thus a clearly negative relation with mortality.
mdl_1 <- lm(mortality ~ insured_rate, data = dt)
print(summary(mdl_1))

##
## Call:
## lm(formula = mortality ~ insured_rate, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -323.18  -82.71   12.61   76.39  390.44
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    670.120     583.196   1.149   0.256
## insured_rate     2.630       6.363   0.413   0.681
##
## Residual standard error: 140.2 on 49 degrees of freedom
## Multiple R-squared:  0.003473,    Adjusted R-squared:  -0.01686
## F-statistic: 0.1708 on 1 and 49 DF,  p-value: 0.6812

# These results are the opposite of what we expected them to be:
# - insurance seems not to be correlated with health and mortality;
# - looking at the scatterplot (with the fitted abline) it even seems that insuran
# ce increases mortality (even if we know the coefficient is not significant).
# Is it possible? Is insurance really useless, or is the model missing something?
```

```
# The result is probably mainly driven by OVB; it is wrong to just consider the insurance effect on crude mortality.  
# Two DISTINCT mechanisms inflate the raw insurance-mortality correlation and must not be conflated:  
# (i) a MECHANICAL age-Medicare channel: the elderly are near-universally covered via Medicare AND have higher  
#     mortality, so any state with an older population mechanically shows both high insurance and high crude mortality;  
# (ii) genuine ADVERSE SELECTION (an asymmetric-information story): individuals with a higher probability of dying  
#     have a stronger incentive to obtain coverage, again tilting the insured pool toward higher-risk individuals.  
# Both push the bivariate coefficient in the "wrong" (positive) direction; in this state-level setting the  
# dominant force is (i), the age-Medicare composition effect.  
  
# Let's continue the analysis, perhaps checking whether the presence of outliers is driving the output.
```

```
boxplot(dt$mortality,  
        main = "Boxplot: crude mortality rate",  
        ylab = "Deaths per 100,000")
```



```
# One value is a possible outlier. Its value is much higher than any other.  
# In practice we may need to take a deeper look at the dataset. Is that value due  
# to an error, or does it come from a state that is underrepresented in our dataset?
```



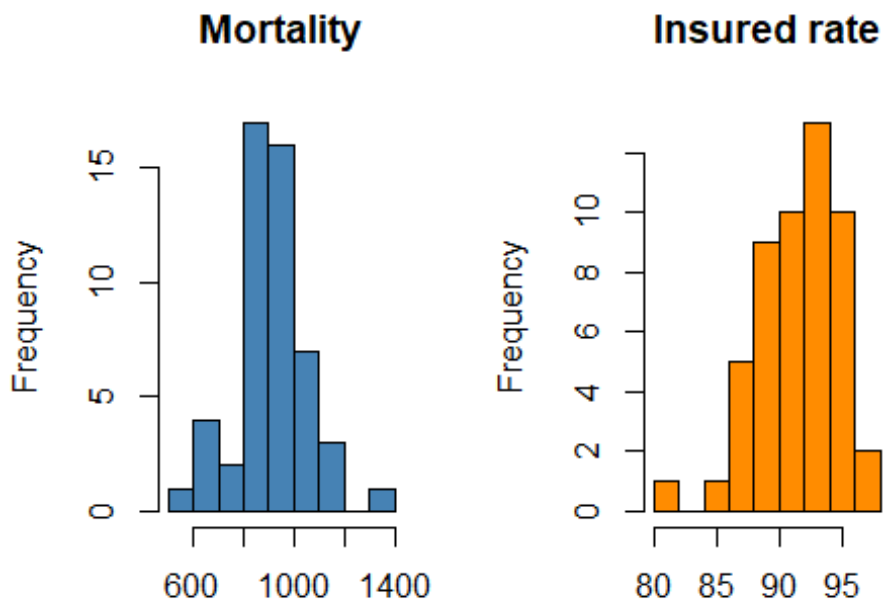
```

dt2 <- subset(dt, mortality <= 1200)
lm_outliers<- lm(mortality ~ insured_rate, data=dt2)
summary(lm_outliers)

##
## Call:
## lm(formula = mortality ~ insured_rate, data = dt2)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -317.14  -77.88   19.43   83.66  217.07
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    790.772     540.722   1.462   0.150
## insured_rate     1.227       5.902   0.208   0.836
##
## Residual standard error: 129.6 on 48 degrees of freedom
## Multiple R-squared:  0.0008992, Adjusted R-squared:  -0.01992
## F-statistic: 0.0432 on 1 and 48 DF,  p-value: 0.8362

# The robustness check confirms that the near-zero and insignificant bivariate
# coefficient is not driven by the high-mortality outlier. Even after removing
# states with mortality > 1200, the OLS estimate on insured_rate remains
# statistically indistinguishable from zero. This is informative: the absence
# of a bivariate signal is a genuine data feature, not a leverage artefact.
par(mfrow = c(1, 2))
hist(dt$mortality, col = "steelblue", main = "Mortality", xlab = "")
hist(dt$insured_rate, col = "darkorange", main = "Insured rate", xlab = "")

```



```
par(mfrow = c(1, 1))
```

```
# The mortality histogram can be approximated graphically by a normal distribution
centred around 900.
```

```
# The insured rate, instead, is LEFT-skewed (a long left tail of low-coverage states),
in accordance with what we saw; the median is 92%.
```

```
# However, in both cases outliers arise: mortality > 1200 and insured rate ≤ 85.
```

```
# In order to see whether our findings point in the right direction, we add some control
variables step by step.
```

```
# The variables chosen are, of course, those that we think should somehow affect mortality.
```

```
# The first variable is age, for the reasoning already explained. Crude mortality
does not account for age – is this an OVB?
```

```
# Of course, we expect states with an older population to report both higher mortality
and a higher level of insurance.
```

```
# ADD AGE CONTROLS (Model 2)
```

```
mdl_2 <- lm(mortality ~ insured_rate + age, data = dt)
```

```
print(summary(mdl_2))
```

```
##
```

```
## Call:
```

```
## lm(formula = mortality ~ insured_rate + age, data = dt)
```

```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -155.08  -84.73  -26.24   94.25  239.61
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -18.721    458.633  -0.041   0.968
## insured_rate    -7.117     5.109  -1.393   0.170
## age           40.770     6.752   6.038 2.19e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 106.8 on 48 degrees of freedom
## Multiple R-squared:  0.4336, Adjusted R-squared:  0.41
## F-statistic: 18.38 on 2 and 48 DF,  p-value: 1.187e-06

# In states with a higher median age, there is a linear marginal increase in mortality equal
# to 40.77 additional deaths per 100,000 for each one-year increase in the state's median age.

# We see that, as expected, age is highly relevant in our analysis.
# However, the probability of dying does not increase linearly with age.
# Reasoning about it, it is unreasonable that the effect of age on mortality for someone turning 20 is the same as for someone turning 90.
mdl_2_quad <- lm(mortality ~ insured_rate + age + I(age^2), data = dt)
print(summary(mdl_2_quad))

##
## Call:
## lm(formula = mortality ~ insured_rate + age + I(age^2), data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -160.41  -86.00  -11.71   86.81  260.24
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2256.172    2188.335  -1.031   0.308
## insured_rate    -7.085     5.104  -1.388   0.172
## age          157.233    111.586   1.409   0.165
## I(age^2)       -1.512     1.446  -1.046   0.301
##
## Residual standard error: 106.7 on 47 degrees of freedom
## Multiple R-squared:  0.4465, Adjusted R-squared:  0.4112
## F-statistic: 12.64 on 3 and 47 DF,  p-value: 3.495e-06

mdl_2.2 <- lm(medical_mortality ~ insured_rate + age + I(age^2), data = dt)
print(summary(mdl_2.2))
```

```
##
## Call:
## lm(formula = medical_mortality ~ insured_rate + age + I(age^2),
##     data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.007  -56.725   -9.304   51.704  170.857
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -1853.088   1472.697  -1.258   0.214
## insured_rate    -5.307     3.435  -1.545   0.129
## age           125.455    75.095   1.671   0.101
## I(age^2)       -1.270     0.973  -1.305   0.198
##
## Residual standard error: 71.8 on 47 degrees of freedom
## Multiple R-squared:  0.4529, Adjusted R-squared:  0.418
## F-statistic: 12.97 on 3 and 47 DF,  p-value: 2.676e-06
```

However, our model is built on a dataset that uses median age; thus, reasoning about personal (individual-level) effects cannot be done.
The results of the model are statistically insignificant, and the coefficient even has a sign opposite to the one expected.

Joint test: is the age contribution (linear + quadratic) significant?
The joint test is the right tool to use.
car::linearHypothesis mdl_2_quad, c("age=0", "I(age^2)=0"))

```
##
## Linear hypothesis test:
## age = 0
## I(age^2) = 0
##
## Model 1: restricted model
## Model 2: mortality ~ insured_rate + age + I(age^2)
##
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      49 963271
## 2      47 535018  2    428253 18.811 9.967e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
car::linearHypothesis(mdl_2.2, c("age=0", "I(age^2)=0"))
```

```
##
## Linear hypothesis test:
## age = 0
## I(age^2) = 0
##
## Model 1: restricted model
## Model 2: medical_mortality ~ insured_rate + age + I(age^2)
```

```
##
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1      49 442120
## 2      47 242308  2    199813 19.379 7.285e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Marginal effect of age in the quadratic specification

```
b2 <- coef mdl_2_quad)
marginal_age <- function(x) b2["age"] + 2 * b2["I(age^2)"] * x
summary(dt$median_age)
```

```
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  31.20  37.70   38.80   38.79  39.85   45.10
```

```
marginal_age(31.20)
```

```
##      age
## 62.89457
```

```
marginal_age(37.70)
```

```
##      age
## 43.24071
```

*# !!the results are not in line with reality: the quadratic age terms are INDIVIDUALLY insignificant ($p=0.17$, $p=0.30$),
yet the IMPLIED marginal effects (+62.9 deaths/100k at median age 31, +43.2 at 38) are economically IMPLAUSIBLE --
far too large, and with implausible curvature, for a one-year change in a STATE-LEVEL median age!!*

For this small age gap, the best approximation is linear, even if economically/socially it is exponential.

#

IMPORTANT – AGGREGATION BIAS / ECOLOGICAL FALLACY:

The apparent weakness of the age effect here is not a true economic finding.

It reflects a fundamental limitation of using state-level medians as regressors.

Median age across U.S. states in 2019 spans roughly 31–44 years – a compressed range of ~13 years. At the individual level, the probability of death rises

sharply and nonlinearly with age (Gompertz law), but that curvature is almost entirely invisible once data are collapsed to state medians.

This is a canonical instance of the ecological fallacy: relationships estimated on aggregate units (states) cannot be assumed to replicate individual-level relationships (Deaton & Lubotsky, 2003; Lynch et al., 2004).

The correct takeaway is not that age does not matter for mortality – it obviously does – but that state-level median age is a poor proxy for the age-composition effect we are trying to control for.

Switching to age-adjusted mortality (as done in Model 5 Later) is precisely the right methodological response: it removes the demographic composition effect at the data construction stage rather than trying to model it imperfectly with a quadratic in median age.

```

# Visual comparison of functional forms
age_grid <- seq(min(dt$age), max(dt$age), length.out = 200)

y_lin <- predict mdl_2,   newdata = data.frame(age = age_grid,
                                                insured_rate = mean(dt$insured_rate))
y_quad <- predict mdl_2_quad, newdata = data.frame(age = age_grid,
                                                    insured_rate = mean(dt$insured_rate)))

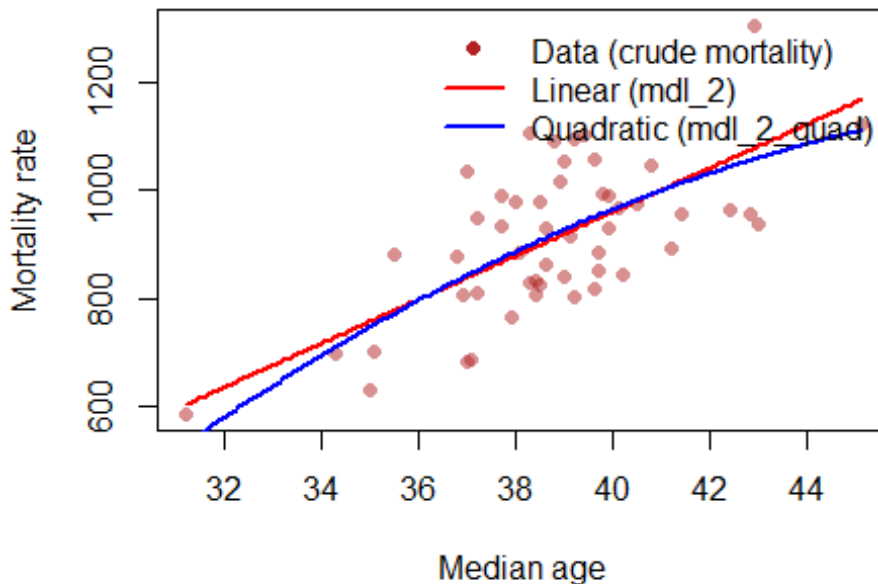
plot(dt$age, dt$mortality,
     pch = 16,
     col = adjustcolor("firebrick", alpha = 0.5),
     xlab = "Median age",
     ylab = "Mortality rate",
     main = "Model comparison: age-mortality")

lines(age_grid, y_lin, col = "red", lwd = 2)
lines(age_grid, y_quad, col = "blue", lwd = 2)

legend("topright",
      legend = c("Data (crude mortality)", "Linear (mdl_2)", "Quadratic (mdl_2_quad)"),
      col = c("firebrick", "red", "blue"),
      pch = c(16, NA, NA),
      lty = c(NA, 1, 1),
      lwd = c(NA, 2, 2),
      bty = "n")

```

Model comparison: age–mortality



*# Graphically, we can note how the quadratic control is wrongly fitting the data;
in addition, the linear model is only noisily fitting the data.
These are red flags on our specification... we can do better.*

ADD POVERTY RATE AND LOG INCOME (Model 3)

```
mdl_3 <- lm(mortality ~ insured_rate + age + poverty_rate + log_income, data = dt)
print(summary(mdl_3))
```

```
##
```

```
## Call:
```

```
## lm(formula = mortality ~ insured_rate + age + poverty_rate +  
##     log_income, data = dt)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -115.38  -39.36    6.10   35.38  127.97
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)  5633.643    963.781   5.845 4.94e-07 ***  
## insured_rate    12.339      3.092   3.991 0.000235 ***  
## age             24.944      3.835   6.505 5.05e-08 ***  
## poverty_rate     2.387      4.771   0.500 0.619177  
## log_income     -618.422    84.246  -7.341 2.82e-09 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 54.36 on 46 degrees of freedom
## Multiple R-squared:  0.8594, Adjusted R-squared:  0.8472
## F-statistic: 70.28 on 4 and 46 DF,  p-value: < 2.2e-16

# FULLY SPECIFIED MODEL WITH INTERACTION (Model 4)
# We now interact c_insured_rate with c_log_income to keep the functional form of
# income consistent throughout the specification while rendering main effects stru
# cturally interpretable.
mdl_4 <- lm(mortality ~ c_insured_rate + age + poverty_rate + c_log_income +
            c_insured_rate:c_log_income, data = dt)
print(summary(mdl_4))

##
## Call:
## lm(formula = mortality ~ c_insured_rate + age + poverty_rate +
##     c_log_income + c_insured_rate:c_log_income, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -113.128  -33.340   -3.821   34.301  113.937
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      16.896     174.897   0.097  0.92347
## c_insured_rate     12.609       2.993   4.212  0.00012 ***
## age              23.870       3.746   6.373 8.71e-08 ***
## poverty_rate     -3.372       5.406  -0.624  0.53596
## c_log_income    -724.182      96.505  -7.504 1.83e-09 ***
## c_insured_rate:c_log_income  39.734     19.431   2.045  0.04674 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

## Residual standard error: 52.57 on 45 degrees of freedom
## Multiple R-squared:  0.8713, Adjusted R-squared:  0.857
## F-statistic: 60.95 on 5 and 45 DF,  p-value: < 2.2e-16

# Visual comparison: mdl_3 vs mdl_4
# Fix all covariates at their mean, vary insured_rate across its observed range
ins_grid <- seq(min(dt$insured_rate), max(dt$insured_rate), length.out = 200)

newdata_base_m3 <- data.frame(
  insured_rate = ins_grid,
  age          = mean(dt$age),
  poverty_rate = mean(dt$poverty_rate),
  log_income   = mean(dt$log_income)
)

newdata_base_m4 <- data.frame(
  c_insured_rate = ins_grid - mean(dt$insured_rate),
  age            = mean(dt$age),
  poverty_rate   = mean(dt$poverty_rate),

```



```

  c_log_income = 0 # Mean-centered baseline is 0
)

y_md13 <- predict(md1_3, newdata = newdata_base_m3)
y_md14 <- predict(md1_4, newdata = newdata_base_m4)

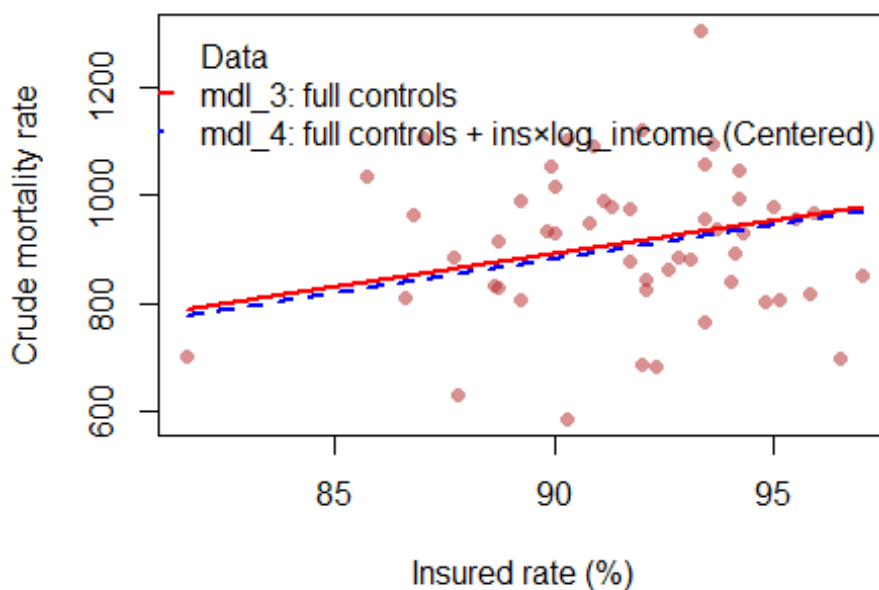
plot(dt$insured_rate, dt$mortality,
     pch = 16,
     col = adjustcolor("firebrick", alpha = 0.5),
     xlab = "Insured rate (%)",
     ylab = "Crude mortality rate",
     main = "Model comparison: mortality ~ insured rate\n(covariates fixed at mean)
")

lines(ins_grid, y_md13, col = "red", lwd = 2)
lines(ins_grid, y_md14, col = "blue", lwd = 2, lty = 2)

legend("topright",
      legend = c("Data",
                  "mdl_3: full controls",
                  "mdl_4: full controls + ins×log_income (Centered)"),
      col = c("firebrick", "red", "blue"),
      pch = c(16, NA, NA),
      lty = c(NA, 1, 2),
      lwd = c(NA, 2, 2),
      bty = "n")

```

Model comparison: mortality ~ insured rate (covariates fixed at mean)



CORRECTION ON THE MODEL TO FIND FINAL CONCLUSION (Model 5)

```
mdl_5 <- lm(agemortality ~ c_insured_rate + poverty_rate + c_log_income +  
            c_insured_rate:c_log_income, data = dt)  
print(summary(mdl_5))  
  
##  
## Call:  
## lm(formula = agemortality ~ c_insured_rate + poverty_rate + c_log_income +  
##     c_insured_rate:c_log_income, data = dt)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -118.197  -22.516    0.794   35.844   81.911   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)      652.379     58.113   11.226 9.06e-15 ***  
## c_insured_rate       2.248       2.534    0.887   0.379      
## poverty_rate        7.142       4.905    1.456   0.152      
## c_log_income      -360.033     82.393   -4.370 7.03e-05 ***  
## c_insured_rate:c_log_income  9.619     18.247    0.527   0.601      
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 49.86 on 46 degrees of freedom  
## Multiple R-squared:  0.6986, Adjusted R-squared:  0.6724   
## F-statistic: 26.65 on 4 and 46 DF,  p-value: 1.792e-11
```

#WHY AGE-ADJUSTED MORTALITY IS THE PREFERRED OUTCOME:

*# Age-adjusted mortality (agemortality) is constructed by the CDC using direct standardisation to a reference population (the 2000 U.S. Standard Population).
It answers the counterfactual question: "what would the mortality rate be if every state had the same age distribution?" By using this outcome we remove the confounding role of age composition from the left-hand side of the regression entirely, rather than attempting to absorb it through a noisy quadratic in median age on the right-hand side.
This choice also directly addresses the adverse selection concern raised in Models 1 and 2: older populations self-select into insurance at higher rates (particularly via Medicare), inflating the bivariate insurance-mortality correlation. Age-standardisation at the outcome level breaks that mechanical link, allowing the insurance coefficient to reflect something closer to the conditional association of interest.
Note that age-adjusted mortality is still an aggregate, ecological measure.*

Visual comparison: mdl_3 vs mdl_4 vs mdl_5

```
ins_grid2 <- seq(min(dt$insured_rate), max(dt$insured_rate), length.out = 200)  
  
newdata_base_m5 <- data.frame(  
  c_insured_rate = ins_grid2 - mean(dt$insured_rate),
```

```

poverty_rate = mean(dt$poverty_rate),
c_log_income = 0
)

y_md13_pt2 <- predict(md1_3, newdata = newdata_base_m3)
y_md14_pt2 <- predict(md1_4, newdata = newdata_base_m4)
y_md15     <- predict(md1_5, newdata = newdata_base_m5)

plot(dt$insured_rate, dt$mortality,
     pch = 16,
     col = adjustcolor("firebrick", alpha = 0.5),
     xlab = "Insured rate (%)",
     ylab = "Mortality rate",
     main = "Model comparison: crude vs age-adjusted mortality\n(covariates fixed
at mean)",
     ylim = range(c(dt$mortality, dt$agemortality, y_md13_pt2, y_md14_pt2, y_md1
5)))

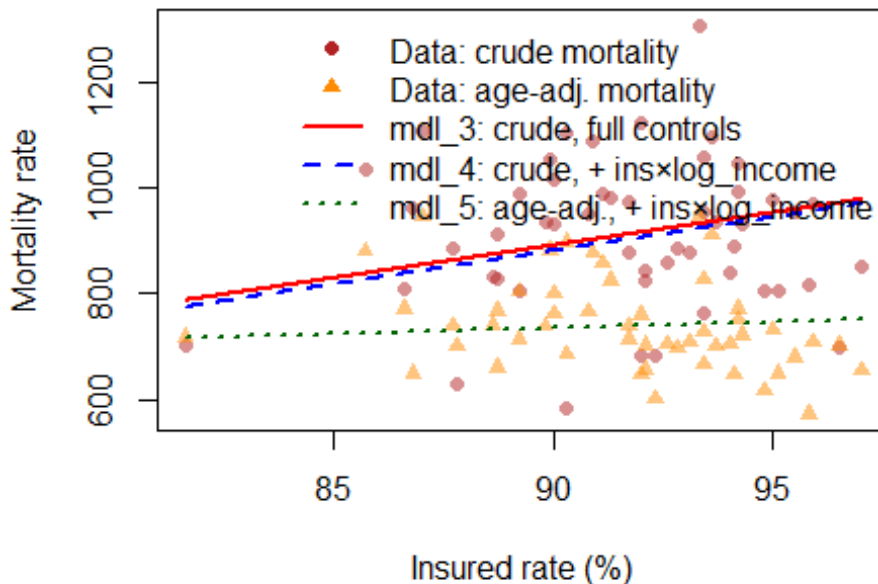
points(dt$insured_rate, dt$agemortality,
       pch = 17,
       col = adjustcolor("darkorange", alpha = 0.5))

lines(ins_grid2, y_md13_pt2, col = "red",      lwd = 2, lty = 1)
lines(ins_grid2, y_md14_pt2, col = "blue",     lwd = 2, lty = 2)
lines(ins_grid2, y_md15,    col = "darkgreen", lwd = 2, lty = 3)

legend("topright",
      legend = c("Data: crude mortality",
                  "Data: age-adj. mortality",
                  "mdl_3: crude, full controls",
                  "mdl_4: crude, + ins×log_income",
                  "mdl_5: age-adj., + ins×log_income"),
      col     = c("firebrick", "darkorange", "red", "blue", "darkgreen"),
      pch     = c(16, 17, NA, NA, NA),
      lty     = c(NA, NA, 1, 2, 3),
      lwd     = c(NA, NA, 2, 2, 2),
      bty     = "n")

```

Model comparison: crude vs age-adjusted mortality (covariates fixed at mean)



From the graphical comparison we see that switching to age-adjusted mortality (green line, mdl_5) FLATTENS the steep positive crude-mortality slopes (red/blue, mdl_3 /mdl_4) down toward zero. Crucially, the mdl_5 slope is still POSITIVE (coefficient on # c_insured_rate = +2.248), NOT negative: age-adjustment removes most of the spurious positive association but does NOT turn it into the protective (negative) relationship theory would predict. The association is essentially flat and, as the regression below shows, statistically indistinguishable from zero ($p = 0.379$).

From the estimated model we see that:

- # - c_insured_rate: the baseline effect of insurance on age-adjusted mortality before accounting for the interaction with income;
- # - c_log_income: captures the independent effect of income on mortality, separate from what is already captured via the interaction;
- # - c_insured_rate:c_log_income: the interaction allows the protective effect of insurance to vary with the income level of a state. A significant coefficient indicates that the relationship between insurance and mortality is moderated by income

Note: OLS standard errors assume homoskedastic errors. With state-level data ($n = 51$), heteroskedasticity is a realistic concern: richer/larger states may have systematically different residual variance. We therefore compute heteroskedasticity-robust standard errors below.

Goodness of fit summaries

```
summary mdl_1$ r.squared
```

```
## [1] 0.003473268
```

```
summary mdl_2$ r.squared
```

```
## [1] 0.4336362
```

```
summary mdl_2_quad$ r.squared
```

```
## [1] 0.4465116
```

```
summary mdl_3$ r.squared
```

```
## [1] 0.8593835
```

```
summary mdl_4$ r.squared
```

```
## [1] 0.8713391
```

```
summary mdl_5$ r.squared
```

```
## [1] 0.6985767
```

```
summary mdl_1$ adj.r.squared
```

```
## [1] -0.01686401
```

```
summary mdl_2$ adj.r.squared
```

```
## [1] 0.4100377
```

```
summary mdl_2_quad$ adj.r.squared
```

```
## [1] 0.4111825
```

```
summary mdl_3$ adj.r.squared
```

```
## [1] 0.847156
```

```
summary mdl_4$ adj.r.squared
```

```
## [1] 0.8570434
```

```
summary mdl_5$ adj.r.squared
```

```
## [1] 0.672366
```

R² PROGRESSION

*# The sharp jump in R² from mdl_1 (~0.003) to mdl_3 (~0.86) confirms that
omitted variable bias was huge in the bivariate specification: once income
and poverty are included, the model accounts for the vast majority of
cross-state variance in mortality.*

```

# However,  $R^2$  alone is not a valid model selection criterion:
# 1)  $R^2$  never decreases when regressors are added, even irrelevant ones.
# Adjusted  $R^2$ , which penalises for additional parameters, is the
# appropriate scalar summary for nested model comparison.
# 2) A high  $R^2$  does not imply correct specification or causal validity.
# 3) Critical: mdl_5's  $R^2$  (0.70) is NOT comparable to that of mdl_1-mdl_4. mdl_5 has
# a different dependent variable (age-adjusted mortality), hence a different Total Sum
# of Squares in the denominator.  $R^2$  is only comparable across models that share the
# same outcome; the drop from 0.87 (mdl_4) to 0.70 (mdl_5) reflects a different
# variance decomposition, NOT a deterioration in fit.

# The preferred model for inference remains mdl_5, selected on theoretical
# grounds (age-adjusted outcome, income interaction) rather than  $R^2$  alone.

# Compute heteroskedasticity-robust standard errors

# NOTE ON HC1 vs HC3 IN SMALL SAMPLES:

# HC1 is the simple degrees-of-freedom rescaling of the classical White (HC0)
# estimator, multiplying the variance by  $n/(n-k)$ ; it is the default "robust" SE
# in many packages (e.g. Stata). HC3 instead reweights each squared residual by
#  $1/(1-h_{ii})^2$ , where  $h_{ii}$  is the leverage of observation  $i$ ; this inflates the
# contribution of high-leverage points and approximates a jackknife / leave-one-
# out variance. The two estimators coincide as  $n$  grows, but can differ noticeably
# in small, high-leverage samples like ours -- hence the choice motivated below.

# Some simulation studies (MacKinnon & White, 1985; Long & Ervin, 2000)
# suggest that HC3 has better finite-sample properties when  $n$  is small or when the
# design matrix contains high-leverage points.
# With 51 observations, a few states (e.g. those with extreme income or
# insurance values) may exert disproportionate leverage.

vcov_robust <- vcovHC(mdl_5, type = "HC3")
robust_se <- coeftest(mdl_5, vcov. = vcov_robust)

summary(mdl_5)

##
## Call:
## lm(formula = agemortality ~ c_insured_rate + poverty_rate + c_log_income +
##     c_insured_rate:c_log_income, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -118.197  -22.516    0.794   35.844   81.911
##

```

```
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      652.379      58.113   11.226 9.06e-15 ***
## c_insured_rate      2.248       2.534    0.887  0.379
## poverty_rate       7.142       4.905    1.456  0.152
## c_log_income     -360.033      82.393   -4.370 7.03e-05 ***
## c_insured_rate:c_log_income  9.619      18.247    0.527  0.601
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.86 on 46 degrees of freedom
## Multiple R-squared:  0.6986, Adjusted R-squared:  0.6724
## F-statistic: 26.65 on 4 and 46 DF, p-value: 1.792e-11
```

robust_se

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      652.3786      75.9259   8.5923 4.023e-11 ***
## c_insured_rate      2.2484       2.7921    0.8053 0.4248087
## poverty_rate       7.1424       6.6179    1.0793 0.2861024
## c_log_income     -360.0329     99.5720  -3.6158 0.0007403 ***
## c_insured_rate:c_log_income  9.6192      21.6443    0.4444 0.6588219
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

We see that:

- OLS coefficient estimates remain identical between the two outputs, as robust standard errors only affect inference (SEs, t-stats, p-values), not the point estimates themselves.

- Under heteroskedasticity the classical OLS SEs are BIASED (not "inflated" in a fixed direction): the bias can go either way, depending on the correlation between the squared errors and the leverage of each observation. In THIS sample, every robust SE is LARGER than its classical counterpart, so the homoskedastic OLS SEs were understating the true sampling uncertainty; the HC3 correction widens the standard errors (and confidence intervals) accordingly.

- The practical implication is that any variable that appeared significant under OLS but turns insignificant under robust SEs should be treated with caution. In our case NO such reversal occurs: c_log_income (and the intercept) remain significant, and the insurance terms remain insignificant, under both classical and robust inference.

In addition to the regression itself, it is interesting to study the single

variables and whether their interaction drives our findings.

Robust joint test: does insurance coverage (direct + interaction) matter?

H0: $\theta_{c_insured_rate} = 0$ AND $\theta_{(c_insured_rate:c_log_income)} = 0$

Testing both together is important because individual t-tests may be

misleading when regressors are correlated (as insured_rate and its

interaction with income likely are).

Robust joint tests (Model 5)

```
car::linearHypothesis mdl_5,  
                      c("c_insured_rate=0", "c_insured_rate:c_log_income=0"),  
                      vcov. = vcov_robust)
```

##

Linear hypothesis test:

c_insured_rate = 0

c_insured_rate:c_log_income = 0

##

Model 1: restricted model

Model 2: agemortality ~ c_insured_rate + poverty_rate + c_log_income +

c_insured_rate:c_log_income

##

Note: Coefficient covariance matrix supplied.

##

	Res.Df	Df	F	Pr(>F)
## 1	48			
## 2	46	2	0.5823	0.5627

Robust joint test: do income-related controls jointly matter?

H0: $\theta_{poverty_rate} = 0$ AND $\theta_{c_log_income} = 0$

Even if individually insignificant (poverty rate), both variables may together capture

meaningful variation in mortality that should not be omitted.

```
car::linearHypothesis mdl_5,  
                      c("poverty_rate=0", "c_log_income=0"),  
                      vcov. = vcov_robust)
```

##

Linear hypothesis test:

poverty_rate = 0

c_log_income = 0

##

Model 1: restricted model

Model 2: agemortality ~ c_insured_rate + poverty_rate + c_log_income +

c_insured_rate:c_log_income

##

Note: Coefficient covariance matrix supplied.

##

	Res.Df	Df	F	Pr(>F)
--	--------	----	---	--------


```
## 1      48
## 2      46  2 35.04 5.675e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# As we see, income has a highly statistically significant joint effect on age-adjusted mortality.

#####
# INTERACTION TERMS EXPLORATION
# All secondary exploratory models have been standardized to utilize mean-centered rates
#####
# INTERACTION TERMS
# We explore three theoretically motivated interactions:
#
# (a) insurance × poverty: Does the effect of insurance differ
# between high- and low-poverty states?
#
# (b) insurance × income: Does the effect of insurance depend on
# the median income level of the state?
#
# (c) insurance × age: Is the mortality-insurance association
# moderated by the age structure of the population?
#
# All interactions are tested for both age-adjusted outcomes.
#####

#####
# Note on specification: consistent with the rationale for Model 5, median age is
# NOT included as a "mute" control in the age-adjusted interaction models below,
# because the age composition has already been removed at the OUTCOME level.
# (It is retained ONLY in model (c), where the insurance x age interaction is the
# explicit object of study.)

# ---- (a) insurance × poverty -----
mdl_int_pov <- lm(agemortality ~ c_insured_rate * c_poverty_rate + c_log_income, data = dt)

print(summary(mdl_int_pov))

##
## Call:
## lm(formula = agemortality ~ c_insured_rate * c_poverty_rate +
##     c_log_income, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -115.57  -21.80   -1.79   34.31   85.55
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      742.5336      7.6675  96.841 < 2e-16 ***
## c_insured_rate      2.1640      2.5616   0.845  0.4026
## c_poverty_rate      8.6981      4.3024   2.022  0.0491 *
## c_log_income     -334.3199     70.2993  -4.756 1.99e-05 ***
## c_insured_rate:c_poverty_rate    0.3293      1.0120   0.325  0.7464
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.95 on 46 degrees of freedom
## Multiple R-squared:  0.6975, Adjusted R-squared:  0.6711
## F-statistic: 26.51 on 4 and 46 DF, p-value: 1.949e-11

print(car::linearHypothesis mdl_int_pov,
      c("c_poverty_rate = 0",
        "c_insured_rate:c_poverty_rate = 0"),
      vcov. = vcovHC(mdl_int_pov, type = "HC3"))

##
## Linear hypothesis test:
## c_poverty_rate = 0
## c_insured_rate:c_poverty_rate = 0
##
## Model 1: restricted model
## Model 2: agemortality ~ c_insured_rate * c_poverty_rate + c_log_income
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F Pr(>F)
## 1      48
## 2      46  2 0.9284 0.4025

pov_lo <- quantile(dt$c_poverty_rate, 0.25)
pov_hi <- quantile(dt$c_poverty_rate, 0.75)

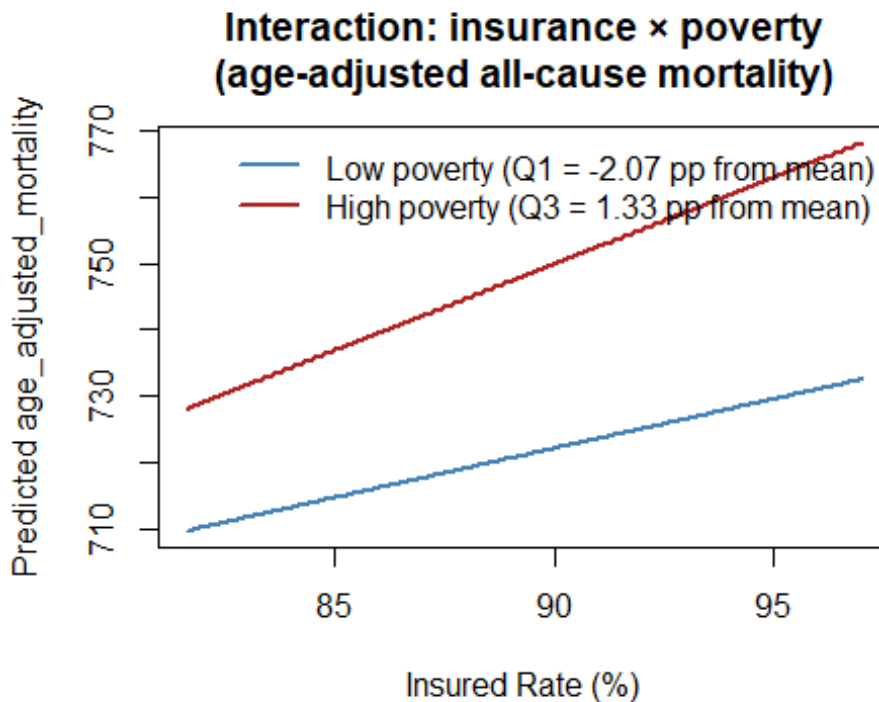
pred_pov_lo <- predict(mdl_int_pov,
      newdata = data.frame(
        c_insured_rate = ins_grid - mean(dt$insured_rate),
        c_poverty_rate = pov_lo,
        c_log_income   = 0)) # mean-centered, so mean = 0; age d
ropped (outcome already age-adjusted)

## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

pred_pov_hi <- predict(mdl_int_pov,
      newdata = data.frame(
        c_insured_rate = ins_grid - mean(dt$insured_rate),
        c_poverty_rate = pov_hi,
        c_log_income   = 0)) # mean-centered, so mean = 0; age d
ropped (outcome already age-adjusted)
```

```
## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

plot(ins_grid, pred_pov_lo,
     type = "l", lwd = 2, col = "steelblue",
     ylim = range(c(pred_pov_lo, pred_pov_hi)),
     xlab = "Insured Rate (%)",
     ylab = "Predicted age_adjusted_mortality",
     main = "Interaction: insurance × poverty\n(age-adjusted all-cause mortality)")
lines(ins_grid, pred_pov_hi, lwd = 2, col = "firebrick")
legend("topright",
      legend = c(sprintf("Low poverty (Q1 = %.2f pp from mean)", pov_lo),
                  sprintf("High poverty (Q3 = %.2f pp from mean)", pov_hi)),
      col = c("steelblue", "firebrick"), lwd = 2, bty = "n")
```



```
# Same for medical mortality
mdl_int_pov_med <- lm(age_adjusted_medical_mortality_rate ~ c_insured_rate * c_poverty_rate + c_log_income, data = dt)
print(summary(mdl_int_pov_med))

##
## Call:
## lm(formula = age_adjusted_medical_mortality_rate ~ c_insured_rate *
##     c_poverty_rate + c_log_income, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -88.688 -14.936   4.243  17.812  64.487
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      493.00191      5.04740   97.674 < 2e-16 ***
## c_insured_rate      1.40676      1.68628    0.834 0.408457
## c_poverty_rate      8.68187      2.83222    3.065 0.003632 **
## c_log_income     -182.94324     46.27677   -3.953 0.000264 ***
## c_insured_rate:c_poverty_rate  -0.04046      0.66621   -0.061 0.951840
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.88 on 46 degrees of freedom
## Multiple R-squared:  0.7093, Adjusted R-squared:  0.684
## F-statistic: 28.06 on 4 and 46 DF, p-value: 7.898e-12

print(car::linearHypothesis mdl_int_pov_med,
      c("c_poverty_rate = 0",
        "c_insured_rate:c_poverty_rate = 0"),
      vcov. = vcovHC(mdl_int_pov_med, type = "HC3"))

##
## Linear hypothesis test:
## c_poverty_rate = 0
## c_insured_rate:c_poverty_rate = 0
##
## Model 1: restricted model
## Model 2: age_adjusted_medical_mortality_rate ~ c_insured_rate * c_poverty_rate
+
+   c_log_income
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F Pr(>F)
## 1      48
## 2      46  2 1.8572 0.1676

# ---- (b) insurance x income -----
mdl_int_inc <- lm(agemortality ~ c_insured_rate * c_log_income + poverty_rate, dat
a = dt)

print(summary(mdl_int_inc))

##
## Call:
## lm(formula = agemortality ~ c_insured_rate * c_log_income + poverty_rate,
##     data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -118.197  -22.516    0.794   35.844   81.911
##
## Coefficients:
```

```
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)      652.379    58.113  11.226 9.06e-15 ***
## c_insured_rate      2.248     2.534   0.887   0.379
## c_log_income     -360.033    82.393  -4.370 7.03e-05 ***
## poverty_rate       7.142     4.905   1.456   0.152
## c_insured_rate:c_log_income  9.619    18.247   0.527   0.601
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.86 on 46 degrees of freedom
## Multiple R-squared:  0.6986, Adjusted R-squared:  0.6724
## F-statistic: 26.65 on 4 and 46 DF, p-value: 1.792e-11

print(car::linearHypothesis mdl_int_inc,
                             c("c_log_income = 0",
                               "c_insured_rate:c_log_income = 0"),
                             vcov. = vcovHC(mdl_int_inc, type = "HC3"))

##
## Linear hypothesis test:
## c_log_income = 0
## c_insured_rate:c_log_income = 0
##
## Model 1: restricted model
## Model 2: agemortality ~ c_insured_rate * c_log_income + poverty_rate
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F    Pr(>F)
## 1      48
## 2      46  2 7.0875 0.002074 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

inc_lo <- quantile(dt$c_log_income, 0.25)
inc_hi <- quantile(dt$c_log_income, 0.75)

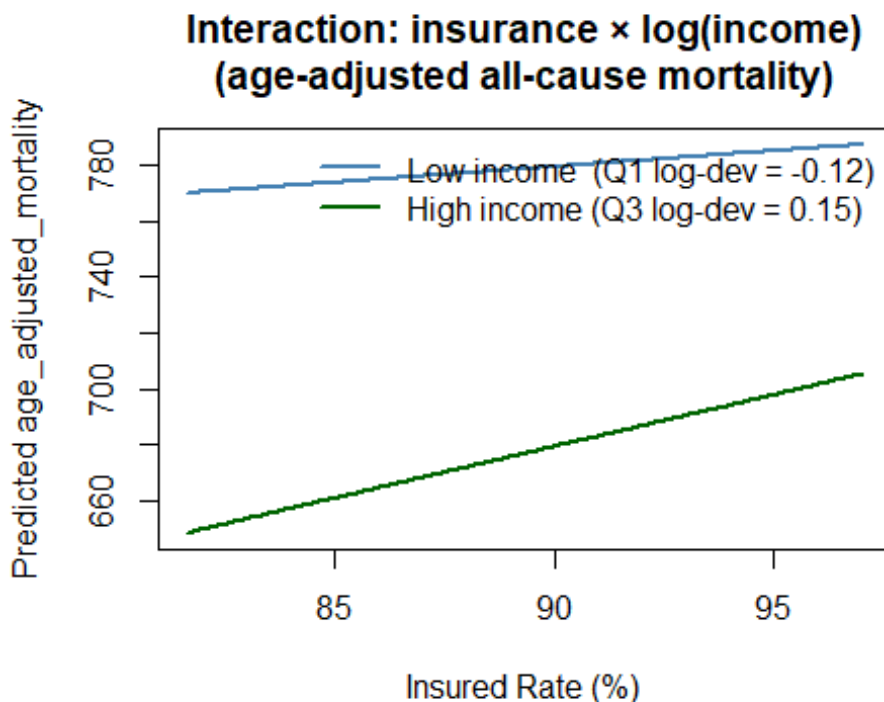
pred_inc_lo <- predict(mdl_int_inc,
                      newdata = data.frame(
                        c_insured_rate = ins_grid - mean(dt$insured_rate),
                        c_log_income   = inc_lo,
                        poverty_rate   = mean(dt$poverty_rate)))

## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

pred_inc_hi <- predict(mdl_int_inc,
                      newdata = data.frame(
                        c_insured_rate = ins_grid - mean(dt$insured_rate),
                        c_log_income   = inc_hi,
                        poverty_rate   = mean(dt$poverty_rate)))
```

```
## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

plot(ins_grid, pred_inc_lo,
     type = "l", lwd = 2, col = "steelblue",
     ylim = range(c(pred_inc_lo, pred_inc_hi)),
     xlab = "Insured Rate (%)",
     ylab = "Predicted age_adjusted_mortality",
     main = "Interaction: insurance × log(income)\n(age-adjusted all-cause mortality)")
lines(ins_grid, pred_inc_hi, lwd = 2, col = "darkgreen")
legend("topright",
      legend = c(sprintf("Low income (Q1 log-dev = %.2f)", inc_lo),
                  sprintf("High income (Q3 log-dev = %.2f)", inc_hi)),
      col = c("steelblue", "darkgreen"), lwd = 2, bty = "n")
```



```
# Same for medical mortality
mdl_int_inc_med <- lm(age_adjusted_medical_mortality_rate ~ c_insured_rate * c_log_income + poverty_rate, data = dt)

print(summary(mdl_int_inc_med))

##
## Call:
## lm(formula = age_adjusted_medical_mortality_rate ~ c_insured_rate *
##     c_log_income + poverty_rate, data = dt)
##
## Residuals:
```

```
##      Min      1Q  Median      3Q      Max
## -87.378 -16.437   3.033  18.760  59.517
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          396.878     38.192  10.392 1.18e-13 ***
## c_insured_rate         1.372       1.665   0.824 0.414183
## c_log_income        -199.090     54.150  -3.677 0.000616 ***
## poverty_rate          7.779       3.224   2.413 0.019858 *
## c_insured_rate:c_log_income  6.843     11.992   0.571 0.571047
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 32.77 on 46 degrees of freedom
## Multiple R-squared:  0.7113, Adjusted R-squared:  0.6862
## F-statistic: 28.34 on 4 and 46 DF,  p-value: 6.746e-12

print(car::linearHypothesis mdl_int_inc_med,
                             c("c_log_income = 0",
                                "c_insured_rate:c_log_income = 0"),
                             vcov. = vcovHC(mdl_int_inc_med, type = "HC3"))

##
## Linear hypothesis test:
## c_log_income = 0
## c_insured_rate:c_log_income = 0
##
## Model 1: restricted model
## Model 2: age_adjusted_medical_mortality_rate ~ c_insured_rate * c_log_income +
##          poverty_rate
##
## Note: Coefficient covariance matrix supplied.
##
##      Res.Df Df       F    Pr(>F)
## 1         48
## 2         46  2 5.5693 0.006824 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

# ---- (c) insurance × age -----
# NOTE ON SPECIFICATION (model c): unlike models (a) and (b), median age (c_age)
# IS retained here -- but NOT as a confounding control. Age-adjusting the outcome
# removes the effect of age COMPOSITION on the average mortality level; it does
# NOT answer whether the EFFECT OF INSURANCE itself differs across states with
# different age structures. That is a question of effect heterogeneity
# (moderation), distinct from confounder control, and it requires age to enter
# the right-hand side as an interaction. The main effect c_age is included only
# because keeping the interaction while omitting its main effect would yield a
# misspecified, non-hierarchical model (violating the marginality principle).
mdl_int_age <- lm(agemortality ~ c_insured_rate * c_age + poverty_rate + c_log_inc
ome, data = dt)
```

```

print(summary mdl_int_age))

##
## Call:
## lm(formula = agemortality ~ c_insured_rate * c_age + poverty_rate +
##     c_log_income, data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -103.571  -26.371    0.466   33.014   87.425
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    640.0167    55.5252   11.527 5.05e-15 ***
## c_insured_rate     4.0142     3.0253    1.327  0.1912
## c_age           -2.9035     3.5270   -0.823  0.4147
## poverty_rate      8.2094     4.4831    1.831  0.0737 .
## c_log_income    -359.8219    77.5798   -4.638 3.04e-05 ***
## c_insured_rate:c_age  0.6949     0.9991    0.696  0.4903
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 49.95 on 45 degrees of freedom
## Multiple R-squared:  0.7041, Adjusted R-squared:  0.6712
## F-statistic: 21.41 on 5 and 45 DF, p-value: 6.674e-11

print(car::linearHypothesis mdl_int_age,
      c("c_age = 0",
        "c_insured_rate:c_age = 0"),
      vcov. = vcovHC mdl_int_age, type = "HC3"))

##
## Linear hypothesis test:
## c_age = 0
## c_insured_rate:c_age = 0
##
## Model 1: restricted model
## Model 2: agemortality ~ c_insured_rate * c_age + poverty_rate + c_log_income
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F Pr(>F)
## 1      47
## 2      45  2 0.1992 0.8201

age_lo <- quantile(dt$c_age, 0.25)
age_hi <- quantile(dt$c_age, 0.75)

pred_age_lo <- predict mdl_int_age,
      newdata = data.frame(
        c_insured_rate = ins_grid - mean(dt$insured_rate),

```



```

        c_age           = age_lo,
        c_log_income    = 0, # c_log_income is mean-centered, so
mean = 0
        poverty_rate    = mean(dt$poverty_rate)))

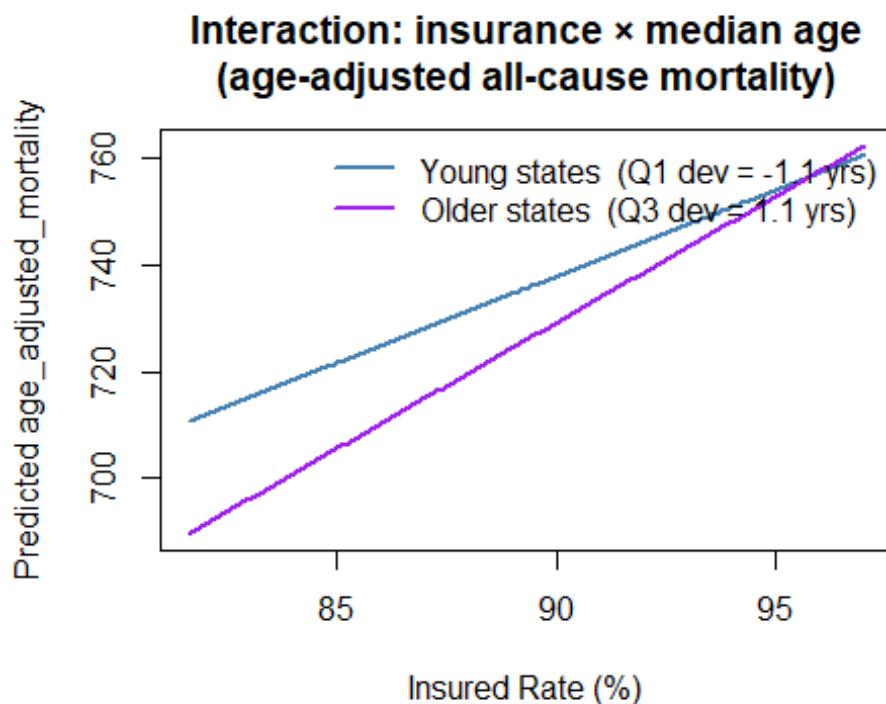
## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

pred_age_hi <- predict mdl_int_age,
                    newdata = data.frame(
                      c_insured_rate = ins_grid - mean(dt$insured_rate),
                      c_age          = age_hi,
                      c_log_income    = 0, # c_log_income is mean-centered, so
mean = 0
                      poverty_rate    = mean(dt$poverty_rate)))

## Warning in data.frame(c_insured_rate = ins_grid - mean(dt$insured_rate), : row
## names were found from a short variable and have been discarded

plot(ins_grid, pred_age_lo,
     type = "l", lwd = 2, col = "steelblue",
     ylim = range(c(pred_age_lo, pred_age_hi)),
     xlab = "Insured Rate (%)",
     ylab = "Predicted age_adjusted_mortality",
     main = "Interaction: insurance x median age\n(age-adjusted all-cause mortalit
y)")
lines(ins_grid, pred_age_hi, lwd = 2, col = "purple")
legend("topright",
      legend = c(sprintf("Young states (Q1 dev = %.1f yrs)", age_lo),
                  sprintf("Older states (Q3 dev = %.1f yrs)", age_hi)),
      col = c("steelblue", "purple"), lwd = 2, bty = "n")

```



```
#####
# V - RESIDUAL DIAGNOSTICS (on mdl_5)
#####

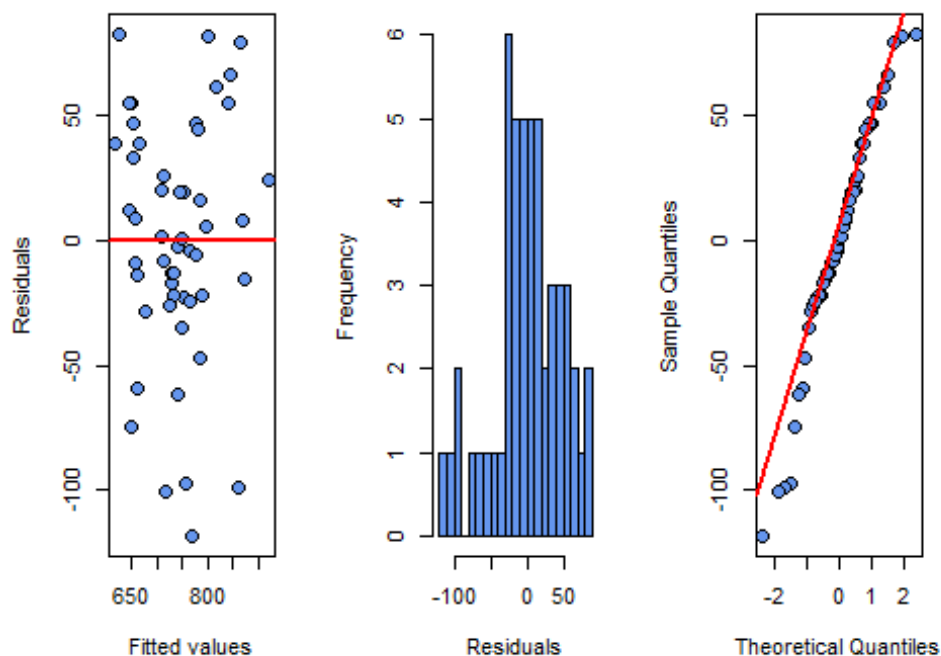
par(mfrow = c(1, 3))

# Residuals vs Fitted
plot(mdl_5$fitted.values, mdl_5$residuals,
     pch = 21, bg = "cornflowerblue", cex = 1.5,
     main = "Residuals vs Fitted (Model 5)",
     xlab = "Fitted values", ylab = "Residuals")
abline(h = 0, col = "red", lwd = 2)

# Histogram of residuals
hist(mdl_5$residuals, col = "cornflowerblue",
     breaks = 15, main = "Residual histogram (Model 5)",
     xlab = "Residuals")

# Q-Q plot
qqnorm(mdl_5$residuals,
       pch = 21, bg = "cornflowerblue", cex = 1.5,
       main = "Q-Q plot of residuals (Model 5)")
qqline(mdl_5$residuals, col = "red", lwd = 2)
```

Residuals vs Fitted (Mod) Residual histogram (Mod) Q plot of residuals (Mod)



```
par(mfrow = c(1, 1))
```

RESIDUAL DIAGNOSTICS – ACADEMIC INTERPRETATION:

#

The three plots above jointly address the core OLS assumptions (Gauss-Markov conditions) and serve as the primary model validation step.

#

(1) RESIDUALS vs FITTED – tests $E(u | X) = 0$ and homoskedasticity

We look for: (i) residuals randomly scattered around zero with no systematic curvature (would indicate functional form misspecification), and (ii) roughly constant spread across the fitted value range (homoskedasticity). A fan-shaped pattern – variance increasing with fitted values – is the canonical heteroskedasticity signature and is common in cross-sectional state-level data where richer/larger states may have more variable mortality outcomes. If such a pattern is visible, it reinforces the choice of HC3 robust SEs made above: OLS estimates remain unbiased (BLUE fails but consistency holds), but classical inference would be invalid without the sandwich correction. With $n = 51$, even a moderate departure from zero-mean residuals at the tails is worth flagging, as it may point to omitted nonlinearities.

#

(2) HISTOGRAM OF RESIDUALS – tests approximate normality

OLS is unbiased and consistent regardless of the residual distribution (the Gauss-Markov theorem requires only zero mean, homoskedasticity, and no serial correlation – not normality). However, finite-sample t- and F-statistics are exact only under normality of the error term. With $n = 51$, the Central Limit Theorem provides only moderate asymptotic comfort. A histogram that is roughly bell-shaped and centred at zero

supports the validity of reported p-values. Heavy tails or multimodality
would suggest either misspecification or the influence of a small number
of high-leverage states driving inference.
#

(3) Q-Q PLOT – the sharpest visual test of normality
Points lying close to the 45-degree reference line indicate that the
empirical quantiles of the residuals match the theoretical normal
quantiles. Deviations in the tails (the first and last few points) are
the most diagnostically informative:
- Upward bow at both ends → heavy tails (leptokurtosis): t-statistics
overstate significance; inference should be treated conservatively.
- S-shape → skewness: may reflect an omitted variable that shifts the
distribution asymmetrically (e.g. a high-mortality state cluster).
With $n = 51$, some tail deviation is expected even from a true normal DGP
and should not be over-interpreted. The Q-Q plot should be read jointly
with the histogram and the residuals-vs-fitted plot.
#

OVERALL CAVEAT ON DIAGNOSTICS WITH $n = 51$:
Formal normality tests have very low power at $n = 51$ and should not be used as t
he
sole basis for accepting or rejecting the normality assumption.
The residual diagnostics do not – and cannot – resolve the fundamental
identification problem: if key confounders are omitted (e.g. racial
composition, healthcare infrastructure quality, urbanisation), the residuals
will absorb their variation, appearing well-behaved despite the model being
misspecified in a causally meaningful sense.

CONNECTING THE EMPIRICAL FINDINGS TO THE THEORETICAL FRAMEWORK:
#

The progression from mdl_1 to mdl_5 documents a textbook sequence of
omitted variable bias correction. The near-zero bivariate coefficient on
insured_rate in mdl_1 ($\beta \approx +2.63$, $p > 0.05$) does not mean insurance is
irrelevant – it means the raw cross-sectional correlation between state
insurance rates and state mortality rates is confounded in multiple,
opposing directions simultaneously:
#

Direction 1 (wrong-sign bias): older states have higher mortality AND
higher insurance enrollment via Medicare. Without controlling for age
composition, insurance appears positively correlated with mortality.
#

Direction 2 (suppressor bias): richer states have lower mortality AND
higher insurance coverage. Income acts as a suppressor variable, pulling
β toward zero in the bivariate case, then revealing an apparently large
positive β once income is explicitly controlled – a sign that the model
is still absorbing income variation through the insurance channel.
#

These two biases partially cancel in mdl_1, producing a near-zero estimate
that could be mistaken for evidence that insurance does not matter. The
progressive model building unmask this cancellation.
#

The final model (mdl_5) on age-adjusted mortality with the income interaction

is our best cross-sectional approximation to the conditional association of
interest. Crucially, even in this preferred specification the coefficient on
`c_insured_rate` remains positive (+2.248) and statistically indistinguishable
from zero ($p = 0.379$); the joint test of insurance (direct effect + income
interaction) is also insignificant (robust $F = 0.58$, $p = 0.56$). In other words,
once age composition is removed at the outcome level and income/poverty are
controlled for, the cross-sectional data do not recover the protective
(negative) association predicted by theory: the point estimate has the
"wrong" (positive) sign and is indistinguishable from no effect.

#

This is not a failure of the analysis but its central substantive finding:
cross-sectional OLS on observational, aggregate state-level data cannot
credibly identify the causal effect of coverage on mortality. The residual
confounding, reverse causality, and aggregation issues discussed below are
severe enough that no sign or magnitude from this design can be given a causal
reading. This is precisely what motivates the move, in Part II, to a research
design with credible exogenous variation in coverage: the Oregon Health
Insurance Experiment (Finkelstein et al., 2012, QJE), a genuine randomized
controlled trial in which a 2008 lottery randomly assigned the opportunity to
apply for Medicaid. "Not distinguishable from zero here" is a statement about
THIS identification strategy, not about the true effect of insurance.

#

REMAINING THREATS TO IDENTIFICATION:

- # (i) Reverse causality: states with unhealthier populations may actively
expand Medicaid coverage in response to high mortality, creating a
positive feedback that attenuates or reverses the causal coefficient.
- # (ii) Unobserved confounders: racial composition, urbanisation, diet,
physical activity, opioid prevalence, and healthcare infrastructure
quality are all correlated with both insurance rates and mortality
and are absent from the model.
- # (iii) Measurement: `insurance_per_100k` captures the share of the population
with any coverage, not the quality or generosity of that coverage.
A state where 92% have catastrophic-only plans is treated identically
to one where 92% have comprehensive plans.

#

WHAT WOULD BE NEEDED FOR CAUSAL IDENTIFICATION

The credible answer requires EXOGENOUS variation in coverage, not more
controls. Part II supplies exactly this through the Oregon Health Insurance
Experiment (Finkelstein et al., 2012, QJE): because the 2008 lottery
randomized the OPPORTUNITY to enroll in Medicaid, winning the lottery is
unconfounded by construction. This yields a clean ITT (the effect of winning
the lottery) and, using the lottery as an instrument for actual enrollment, a
LATE via 2SLS (the effect of coverage on compliers) – the two estimands that
observational OLS cannot recover here. Broader quasi-experimental evidence
points the same way: Sommers, Baicker & Epstein (2012, NEJM) exploit pre/post
state Medicaid expansions, and Goldin, Lurie & McCubbin (2021, QJE) a
randomized IRS outreach experiment; both find significant mortality reductions
that our cross-sectional OLS can only gesture toward but cannot estimate.

#

Crucially, Part II does NOT overturn the mortality finding of Part I – it
reframes it. Under the Oregon randomization the ITT on survival is also

statistically indistinguishable from zero (ITT on the death scale ~ -0.0003 ,
clustered SE ~ 0.0007 , $p \sim 0.64$): a precise null. But the reason is now the
OPPOSITE of Part I's. Here it is not a failure of identification (the design
is clean) but a lack of statistical power: death is a rare event ($\sim 0.8\%$ over a
~ 16 -month horizon), so even a clean experiment cannot pin down a mortality
effect. The same outcome is "indistinguishable from zero" in both parts – in
Part I because the design cannot identify the effect, in Part II because the
experiment cannot detect it. Where the OHIE IS powered (financial strain,
healthcare utilization, mental health) it recovers large, statistically
significant causal effects – demonstrating what credible identification
delivers once power is not the binding constraint.

OLS coefficient estimates remain unchanged by robust SE corrections, but
inference is adjusted to remain valid under heteroskedasticity.
Even flexible specifications, interaction terms, and robust inference do
not automatically confer causal interpretation.
Causal interpretation still requires exogeneity: $E(u \mid X) = 0$.

Part II

2026-06-05

```
#####
#####
##### Part II - Causal Inference (RCT) #####
##### Oregon Health Insurance Experiment #####
#####
#####
###
### Research question: what is the causal effect of Medicaid coverage on health-care use, financial strain, and health?
###
### Design: the 2008 Oregon Lottery randomly selected adults to be allowed to apply for Medicaid -> a real randomized experiment (RCT).
###
### Main method: Intention-to-Treat (ITT) = difference in mean outcomes between lottery winners and losers (the estimand is the effect of the offer of coverage, identified by randomization alone).
###
### Data: OHIE Public Use Files, Harvard Dataverse, DOI 10.7910/DVN/SJG1ED.
### Reference paper: Finkelstein et al. (2012), QJE 127(3): 1057-1106.
###
### Note on sources: commands taught in the course are used natively.
### Commands not covered in the course are flagged explicitly with their external source.
###
#####

# ---- Packages -----
library(ggplot2)    # all figures
library(lmtest)     # coeftest

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric

library(sandwich)    # vcovHC (Lab1) + vcovCL
library(MatchIt)     # matchit / match_data
library(cobalt)      # Love.plot

## cobalt (Version 4.6.3, Build Date: 2026-05-29)

##
## Attaching package: 'cobalt'
```



```

## The following object is masked from 'package:MatchIt':
##
##      lalonde

library(car)          # LinearHypothesis

## Loading required package: carData

dt <- read.csv("dt.csv")
# Survey analysis sample: 12m survey responders with a non-zero survey weight
# (resp12 is built in the data-prep script as sample_12m_resp == 1 & weight_12m !=
# 0).
da <- subset(dt, resp12)  # da = survey analysis sample

# Pre-randomization covariates (recorded on the Lottery list, before the draw)
baseline <- c("birthyear_list", "female_list", "english_list", "self_list",
              "first_day_list", "have_phone_list", "pobox_list", "zip_msa")

# -----
# HELPER: clustered standard errors (EXTENSION beyond the Labs)
# -----
# Here the treatment is assigned at the HOUSEHOLD level, so observations within
# a household are not independent and we must cluster the SE on household_id.
# vcovCL() is the clustered analogue of vcovHC(), from the SAME sandwich package.
# Source: Zeileis, Koll & Graham (2020), "Various Versatile Variances: An
#         Object-Oriented Implementation of Clustered Covariances in R", JSS 95(1)
#         (sandwich::vcovCL). Methodology: Cameron & Miller (2015), J. Human
#         Resources 50(2).
#
# vcl(): returns the clustered covariance, but first aligns the cluster vector to
# the rows lm actually used. lm() drops rows with NA (na.omit), so without this
# alignment vcovCL() errors (cluster length != #obs) or mis-clusters whenever a
# variable has missing values (true here: e.g. female_list, zip_msa, and most 12m
# survey outcomes have NAs). Every call below passes the household_id of the exact
# data frame the model was fit on, so the index alignment is correct.
vcl <- function(model, cluster) {
  if (!is.null(model$na.action)) cluster <- cluster[-as.integer(model$na.action)]
  vcovCL(model, cluster = cluster, type = "HC1")
}
clx <- function(model, cluster) {
  coeftest(model, vcov. = vcl(model, cluster))
}
#####
# I - RESEARCH DESIGN VALIDATION
#####
# "RCT" is not a magic word: before trusting the Lottery we must SHOW, on the
# data, that it behaved like an experiment. Three checks: (1A) first stage,
# (1B) balance, (1C) differential non-response.

# ---- 1A. FIRST STAGE: did winning the Lottery change Medicaid enrollment? ----
# The whole experiment hinges on this. If winning did not actually raise

```



```

# coverage, the offer would be a treatment in name only and the ITT would
# measure the effect of nothing. So before anything else we check that the
# instrument bites: we regress actual enrollment on the lottery win.

# Model:  $ohp\_admin = b_0 + b_1 * treatment + (draw) + (hh-size) + u$ 
# The two sets simply reproduce HOW the lottery was run:
# - factor(draw_lottery) = lottery-draw fixed effects
# - factor(numhh_list) = household-size fixed effects
# Enrollment variable here = ohp_admin (full-sample administrative "ever enrolled",
# = ohp_all_ever_matchn_30sep2009). The LATE in Section II.4 instead instruments
# `insured` (= ohp_survey = ohp_all_ever_firstn_30sep2009), the survey-sample
# measure whose first stage is ~0.290. Using two different enrollment measures -
# full-sample first stage (~0.256) vs survey-sample LATE is intentional and
# follows the note to QJE Table III.
fs_full <- lm(ohp_admin ~ treatment + factor(draw_lottery) + factor(numhh_list),
              data = dt)
summary(fs_full)

##
## Call:
## lm(formula = ohp_admin ~ treatment + factor(draw_lottery) + factor(numhh_list),
##     data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.43139 -0.17553 -0.14035 -0.09406  0.98425
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.175529   0.004890  35.898 < 2e-16 ***
## treatment      0.255866   0.003115  82.136 < 2e-16 ***
## factor(draw_lottery)2 -0.008919   0.006601  -1.351 0.176654
## factor(draw_lottery)3 -0.014928   0.006600  -2.262 0.023704 *
## factor(draw_lottery)4 -0.019785   0.006640  -2.980 0.002888 **
## factor(draw_lottery)5 -0.023999   0.006658  -3.605 0.000313 ***
## factor(draw_lottery)6 -0.035184   0.005752  -6.117 9.57e-10 ***
## factor(draw_lottery)7 -0.043558   0.005775  -7.543 4.64e-14 ***
## factor(draw_lottery)8 -0.057691   0.006672  -8.647 < 2e-16 ***
## factor(numhh_list)2  -0.037907   0.003626 -10.455 < 2e-16 ***
## factor(numhh_list)3  -0.124599   0.032671  -3.814 0.000137 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4089 on 74911 degrees of freedom
## Multiple R-squared:  0.08473,    Adjusted R-squared:  0.08461
## F-statistic: 693.5 on 10 and 74911 DF,  p-value: < 2.2e-16

print(clx(fs_full, dt$household_id)["treatment", , drop = FALSE])

```

```
##           Estimate Std. Error  t value Pr(>|t|)
## treatment 0.2558658 0.00350915 72.91389      0

# We get b1 ~ 0.256: winning the lottery raises the probability of ever being on
# Medicaid by about 25 percentage points.
# This reproduces the published QJE first stage, so the "offer" is a strong, real
# treatment. Note b1 is well below 1: not every winner enrolls, and some losers
# get Medicaid anyway. That imperfect take-up is exactly why the offer (ITT) and
# the effect of enrollment (LATE) are different numbers.

# ---- 1B. BALANCE: were winners and losers comparable before the draw? ----
# Randomization implies the two arms should not differ on pre-treatment
# covariates. We test each covariate (lm of the covariate on treatment + hh-size
# FE) and report the difference.
bal_tab <- data.frame()
for (v in baseline) {
  m <- lm(as.formula(paste(v, "~ treatment + factor(numhh_list)")), data = dt)
  ct <- clx(m, dt$household_id)["treatment", ]
  bal_tab <- rbind(bal_tab, data.frame(
    covariate      = v,
    control_mean   = round(mean(dt[[v]][dt$treatment == 0], na.rm = TRUE), 4),
    diff           = round(ct["Estimate"], 4),
    se             = round(ct["Std. Error"], 4),
    p              = round(ct["Pr(>|t|)"], 3),
    row.names      = NULL))
}
print(bal_tab)

##           covariate control_mean    diff    se    p
## 1  birthyear_list    1967.9981  0.1619 0.0995 0.104
## 2   female_list      0.5571 -0.0069 0.0033 0.039
## 3  english_list      0.9220  0.0024 0.0026 0.346
## 4   self_list       0.9180  0.0003 0.0003 0.273
## 5 first_day_list      0.0927  0.0012 0.0025 0.627
## 6 have_phone_list     0.8616 -0.0029 0.0028 0.300
## 7   pobox_list       0.1166  0.0004 0.0027 0.873
## 8      zip_msa       0.7731 -0.0024 0.0036 0.500

# Omnibus joint F-test: regress treatment on all baseline covariates at once and
# test that they are jointly zero. This is our addition on top of the per-row test.
omni <- lm(as.formula(paste("treatment ~", paste(baseline, collapse = " + "),
  "+ factor(numhh_list)")), data = dt)
summary(omni)

##
## Call:
## lm(formula = as.formula(paste("treatment ~", paste(baseline,
## collapse = " + "), "+ factor(numhh_list)")), data = dt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8958 -0.3476 -0.3398  0.6463  0.6722
```

```
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -0.1636565  0.2842299  -0.576   0.5648
## birthyear_list  0.0002587  0.0001442   1.794   0.0728 .
## female_list    -0.0064156  0.0035584  -1.803   0.0714 .
## english_list    0.0066742  0.0064142   1.041   0.2981
## self_list       0.0029505  0.0073351   0.402   0.6875
## first_day_list  0.0035161  0.0060254   0.584   0.5595
## have_phone_list -0.0051141  0.0051320  -0.997   0.3190
## pobox_list      0.0002536  0.0056696   0.045   0.9643
## zip_msa         -0.0032076  0.0043304  -0.741   0.4589
## factor(numhh_list)2  0.2296682  0.0056042  40.981  <2e-16 ***
## factor(numhh_list)3  0.5436765  0.0386483  14.067  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4796 on 74908 degrees of freedom
## (3 observations deleted due to missingness)
## Multiple R-squared:  0.04049,    Adjusted R-squared:  0.04036
## F-statistic: 316.1 on 10 and 74908 DF,  p-value: < 2.2e-16

print(car::linearHypothesis(omni, paste0(baseline, " = 0"),
                                vcov. = vcl(omni, dt$household_id)))

##
## Linear hypothesis test:
## birthyear_list = 0
## female_list = 0
## english_list = 0
## self_list = 0
## first_day_list = 0
## have_phone_list = 0
## pobox_list = 0
## zip_msa = 0
##
## Model 1: restricted model
## Model 2: treatment ~ birthyear_list + female_list + english_list + self_list +
##          first_day_list + have_phone_list + pobox_list + zip_msa +
##          factor(numhh_list)
##
## Note: Coefficient covariance matrix supplied.
##
##   Res.Df Df    F Pr(>F)
## 1   74916
## 2   74908  8 1.3105 0.2327

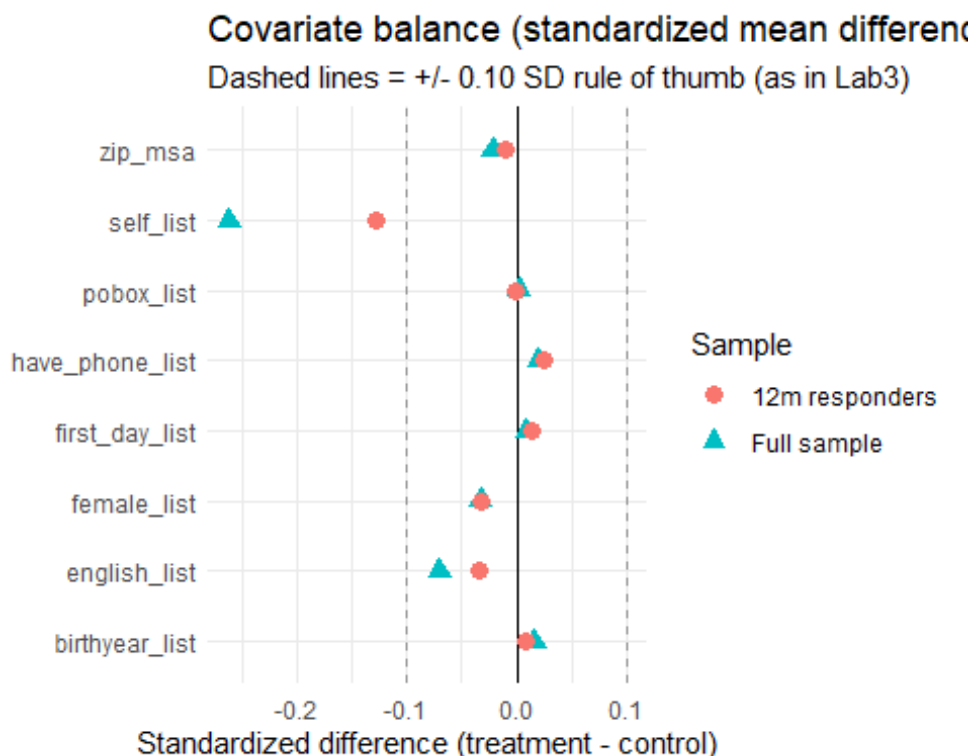
# A non-significant F means we cannot reject balance.

# Visual: a Love plot of standardized mean differences, full sample vs
# responders.
smd <- function(data, v) {
```

```

t <- data[[v]][data$treatment == 1]; c <- data[[v]][data$treatment == 0]
(mean(t, na.rm=TRUE) - mean(c, na.rm=TRUE)) /
  sqrt((var(t, na.rm=TRUE) + var(c, na.rm=TRUE)) / 2)
}
love_df <- rbind(
  data.frame(Covariate = baseline, SMD = sapply(baseline, function(v) smd(dt, v)),
    Sample = "Full sample"),
  data.frame(Covariate = baseline, SMD = sapply(baseline, function(v) smd(da, v)),
    Sample = "12m responders"))
ggplot(love_df, aes(x = SMD, y = Covariate, colour = Sample, shape = Sample)) +
  geom_vline(xintercept = 0) +
  geom_vline(xintercept = c(-0.1, 0.1), linetype = "dashed", colour = "grey60") +
  geom_point(size = 3) +
  labs(title = "Covariate balance (standardized mean differences)",
    subtitle = "Dashed lines = +/- 0.10 SD rule of thumb (as in Lab3)",
    x = "Standardized difference (treatment - control)", y = NULL) +
  theme_minimal()

```



How to read it: each covariate has two points, one per sample (full vs 12m
 # responders); dashed lines mark the +/- 0.10 SD rule of thumb. Almost everything
 # hugs zero in both samples.
 # The one exception is self_list (~ -0.25 SD full, ~ -0.13 responders): it sits
 # outside the band, but this is an artefact of an unconditional SMD,
 # not a broken randomization. self_list is mechanically tied to
 # household size (in 1-person households you self-enrol; in larger ones one member
 # signs up the rest), and household size drives the treatment probability, so a
 # raw, unconditional comparison conflates the two. The authoritative balance check
 # is the regression table + omnibus F just above, which condition on

```

# factor(numhh_list). There self_list is balanced (diff +0.0003, p = 0.273).

# ---- 1C. DIFFERENTIAL NON-RESPONSE: does the survey reintroduce selection? ---
# Outcomes come from a voluntary survey. If winning changed who answers, the
# respondent subsample is no longer randomized. We test whether treatment
# predicts responding (among those mailed the survey).
nonresp <- lm(returned_12m ~ treatment + factor(numhh_list),
               data = subset(dt, sample_12m == 1))
summary(nonresp)

##
## Call:
## lm(formula = returned_12m ~ treatment + factor(numhh_list), data = subset(dt,
##   sample_12m == 1))
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.4281 -0.4110 -0.3937  0.5890  0.6278
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.411045   0.003103  132.461 < 2e-16 ***
## treatment      -0.017311   0.004084   -4.239 2.25e-05 ***
## factor(numhh_list)2  0.017013   0.004511    3.772 0.000162 ***
## factor(numhh_list)3 -0.021550   0.039561   -0.545 0.585944
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4912 on 58401 degrees of freedom
## Multiple R-squared:  0.0005194, Adjusted R-squared:  0.0004681
## F-statistic: 10.12 on 3 and 58401 DF, p-value: 1.17e-06

print(clx(nonresp, subset(dt, sample_12m == 1)$household_id)["treatment", , drop =
  FALSE])

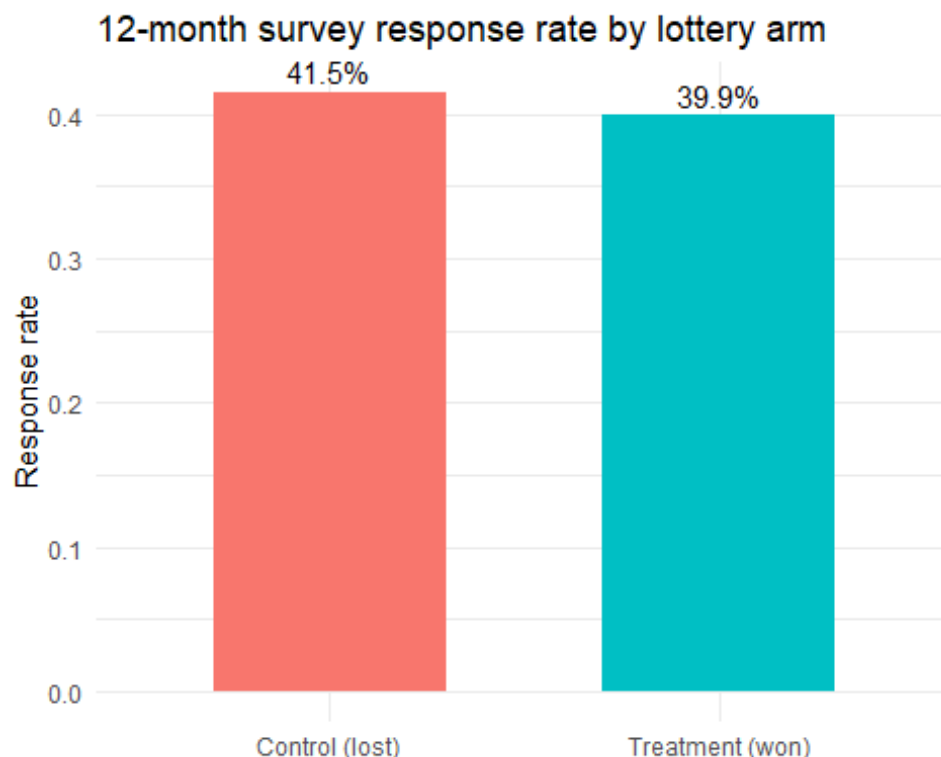
##              Estimate Std. Error  t value      Pr(>|t|)
## treatment -0.01731144 0.004440546 -3.898494 9.690068e-05

# We hoped for ~ 0: if treatment did not move response, answering would be "as
# good as random" given the design, and the respondent subsample would inherit the
# experimental balance untouched. Instead the gap is a small negative number:
# winners respond ~1.7 pp LESS. It is significant only because N is huge.
# So a differential response does exist, and we do not wave it away just
# because it is small. This is a limitation of the model
# that can re-introduce the threat of selection bias on observables when we move
# from the offer to enrollment (non-compliers).

# Visual: response rate by Lottery arm.
rr <- aggregate(returned_12m ~ treatment, data = subset(dt, sample_12m == 1), FUN
= mean)
rr$Arm <- ifelse(rr$treatment == 1, "Treatment (won)", "Control (lost)")
ggplot(rr, aes(x = Arm, y = returned_12m, fill = Arm)) +

```

```
geom_col(width = 0.6, show.legend = FALSE) +
geom_text(aes(label = sprintf("%.1f%%", 100 * returned_12m)), vjust = -0.4) +
labs(title = "12-month survey response rate by lottery arm",
     y = "Response rate", x = NULL) +
theme_minimal()
```



*# How to read it: the two bars are almost the same height (~41.5% control vs
~39.9% treatment). The small gap matches the tiny negative coefficient in the
regression just above: winning nudges response down by less than 2 points.
Small but real, which is why we lean on the survey weights and still flag
residual selection among responders as a limitation.*

```
#####
# II - SELECTION BIAS: why we need the experiment
#####
# On the same data we estimate the effect three ways
# and watch the observational answer fail:
# (1) Naive OLS      : compare insured vs uninsured, "controlling" for demogra
# phics
# (2) Matching       : pair similar units (still blind to unobservables)
# (3) ITT            : compare lottery winners vs losers
# (4) LATE/2SLS      : effect of enrollment for compliers
#
# Estimand note: OLS, Matching and LATE estimate the effect of
# enrollment; the ITT estimates the effect of the offer (~ LATE x first stage).
# So the valid selection-bias contrast is OLS/Matching vs LATE.

# Demographic controls that are plausibly pre-determined (used by OLS/Matching).
```

```
dem <- "age12 + female_list + english_list + race_white_12m + race_black_12m + race_hisp_12m + factor(edu_12m)"
```

```
## ---- (1) NAIVE OLS (observational, biased) -----
```

```
# Model: outcome =  $b_0 + b_1 \text{insured} + (\text{demographics}) + u$   
# We compare the insured and the uninsured, throwing in demographic controls and  
# hoping that is enough. The estimand is the effect of enrollment.
```

```
ols_h <- lm(as.formula(paste("good_health ~ insured +", dem)), data = da)  
summary(ols_h)
```

```
##
```

```
## Call:
```

```
## lm(formula = as.formula(paste("good_health ~ insured +",  
## dem)), data = da)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max  
## -0.9353 -0.5050  0.2620  0.4235  0.7573
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)    0.7893581   0.0206883   38.155 <2e-16 ***  
## insured       -0.0077554   0.0071441   -1.086  0.2777  
## age12         -0.0079253   0.0002655  -29.849 <2e-16 ***  
## female_list   -0.0013578   0.0065274   -0.208  0.8352  
## english_list   0.0033818   0.0150365    0.225  0.8221  
## race_white_12m -0.0167360   0.0107875   -1.551  0.1208  
## race_black_12m -0.0489464   0.0192957   -2.537  0.0112 *  
## race_hisp_12m  -0.0215245   0.0131357   -1.639  0.1013  
## factor(edu_12m)2  0.1186420   0.0093465   12.694 <2e-16 ***  
## factor(edu_12m)3  0.1850507   0.0107366   17.236 <2e-16 ***  
## factor(edu_12m)4  0.3044359   0.0125384   24.280 <2e-16 ***
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 0.4786 on 22470 degrees of freedom
```

```
## (1260 observations deleted due to missingness)
```

```
## Multiple R-squared:  0.06458, Adjusted R-squared:  0.06416
```

```
## F-statistic: 155.1 on 10 and 22470 DF, p-value: < 2.2e-16
```

```
print(clx(ols_h, da$household_id)["insured", , drop = FALSE]) # good health
```

```
##              Estimate Std. Error  t value Pr(>|t|)  
## insured -0.007755445  0.007262123  -1.067931 0.2855632
```

```
ols_d <- lm(as.formula(paste("cost_any_owe_12m ~ insured +", dem)), data = da)  
summary(ols_d)
```

```
##
```

```
## Call:
```

```
## lm(formula = as.formula(paste("cost_any_owe_12m ~ insured +",  
## dem)), data = da)
```



```
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.8371 -0.5570  0.3518  0.4284  0.9371
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.3229325   0.0210601  15.334 < 2e-16 ***
## insured        -0.0616557   0.0072709  -8.480 < 2e-16 ***
## age12          -0.0004253   0.0002705  -1.572  0.11590
## female_list     0.0741058   0.0066453  11.152 < 2e-16 ***
## english_list    0.2165421   0.0152934  14.159 < 2e-16 ***
## race_white_12m  0.0799776   0.0109740   7.288 3.25e-13 ***
## race_black_12m  0.1329393   0.0196145   6.778 1.25e-11 ***
## race_hisp_12m   0.0562834   0.0133757   4.208 2.59e-05 ***
## factor(edu_12m)2 -0.0416537   0.0094977  -4.386 1.16e-05 ***
## factor(edu_12m)3 -0.0299149   0.0109122  -2.741  0.00612 **
## factor(edu_12m)4 -0.1715495   0.0127492 -13.456 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.488 on 22547 degrees of freedom
## (1183 observations deleted due to missingness)
## Multiple R-squared:  0.03342,    Adjusted R-squared:  0.03299
## F-statistic: 77.96 on 10 and 22547 DF,  p-value: < 2.2e-16

print(clx(ols_d, da$household_id)["insured", , drop = FALSE]) # owe medical debt

##              Estimate Std. Error t value    Pr(>|t|)
## insured -0.06165571 0.007574328 -8.14009 4.152535e-16

# Reading: on good_health the coefficient is basically 0 (and not significant),
# observationally, the insured look no healthier than the uninsured.

## ---- (2) MATCHING -----
# Goal of matching is balance.
# Note: Matching is our addition. It is not in the QJE paper. We include it to
# show that even a "good" observational method cannot fix unobserved selection.
dm <- na.omit(da[, c("insured", "good_health", "cost_any_owe_12m", "age12", "female_list",
                    "english_list", "race_white_12m", "race_black_12m", "race_hisp_12m",
                    "edu_12m", "household_id")])
# (a) imbalance before matching
m_out0 <- matchit(insured ~ age12 + female_list + english_list +
                  race_white_12m + race_black_12m + race_hisp_12m + factor(edu_12m),
                  data = dm, method = NULL)
summary(m_out0) # Large Std. Mean Diff. = treated/control not comparable

##
## Call:
```



```
## matchit(formula = insured ~ age12 + female_list + english_list +
##         race_white_12m + race_black_12m + race_hisp_12m + factor(edu_12m),
##         data = dm, method = NULL)
##
## Summary of Balance for All Data:
##               Means Treated Means Control Std. Mean Diff. Var. Ratio
## distance           0.2868           0.2771           0.2224           0.9975
## age12              41.1421           42.4436          -0.1084           0.9711
## female_list         0.6318           0.5774           0.1129              .
## english_list        0.9261           0.9143           0.0451              .
## race_white_12m      0.8394           0.8231           0.0444              .
## race_black_12m      0.0386           0.0323           0.0327              .
## race_hisp_12m       0.1098           0.1193          -0.0305              .
## `factor(edu_12m)`1  0.1827           0.1621           0.0532              .
## `factor(edu_12m)`2  0.5143           0.4896           0.0494              .
## `factor(edu_12m)`3  0.2158           0.2236          -0.0188              .
## `factor(edu_12m)`4  0.0872           0.1247          -0.1331              .
```

```
##               eCDF Mean eCDF Max
## distance           0.0527  0.0907
## age12              0.0296  0.0520
## female_list        0.0544  0.0544
## english_list       0.0118  0.0118
## race_white_12m     0.0163  0.0163
## race_black_12m     0.0063  0.0063
## race_hisp_12m      0.0095  0.0095
## `factor(edu_12m)`1 0.0206  0.0206
## `factor(edu_12m)`2 0.0247  0.0247
## `factor(edu_12m)`3 0.0077  0.0077
## `factor(edu_12m)`4 0.0375  0.0375
```

```
##
## Sample Sizes:
##           Control Treated
## All          16063    6241
## Matched      16063    6241
## Unmatched         0         0
## Discarded         0         0
```

(b) 1:1 nearest-neighbour PS matching

```
m_out1 <- matchit(insured ~ age12 + female_list + english_list +
                  race_white_12m + race_black_12m + race_hisp_12m + factor(edu_1
2m),
```

```
                  data = dm, method = "nearest", link = "logit",
                  distance = "glm", estimand = "ATT")
```

```
summary(m_out1)
```

```
##
## Call:
## matchit(formula = insured ~ age12 + female_list + english_list +
##         race_white_12m + race_black_12m + race_hisp_12m + factor(edu_12m),
##         data = dm, method = "nearest", distance = "glm", link = "logit",
##         estimand = "ATT")
##
```

```

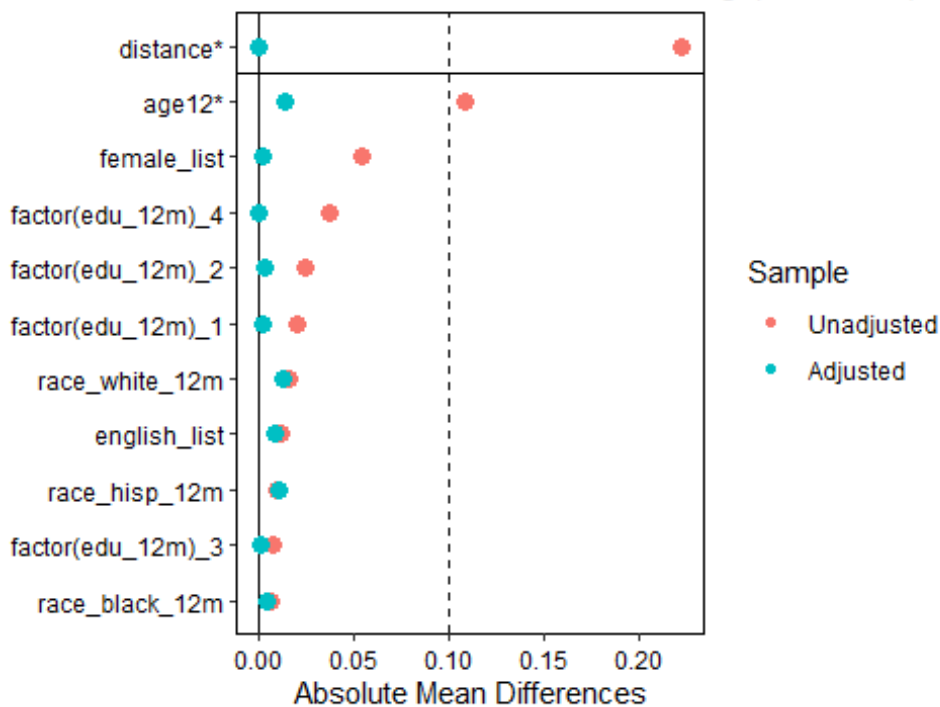
## Summary of Balance for All Data:
##               Means Treated Means Control Std. Mean Diff. Var. Ratio
## distance              0.2868          0.2771          0.2224      0.9975
## age12                 41.1421         42.4436         -0.1084      0.9711
## female_list           0.6318          0.5774          0.1129          .
## english_list          0.9261          0.9143          0.0451          .
## race_white_12m        0.8394          0.8231          0.0444          .
## race_black_12m        0.0386          0.0323          0.0327          .
## race_hisp_12m         0.1098          0.1193         -0.0305          .
## `factor(edu_12m)`1    0.1827          0.1621          0.0532          .
## `factor(edu_12m)`2    0.5143          0.4896          0.0494          .
## `factor(edu_12m)`3    0.2158          0.2236         -0.0188          .
## `factor(edu_12m)`4    0.0872          0.1247         -0.1331          .
##               eCDF Mean eCDF Max
## distance              0.0527      0.0907
## age12                 0.0296      0.0520
## female_list           0.0544      0.0544
## english_list          0.0118      0.0118
## race_white_12m        0.0163      0.0163
## race_black_12m        0.0063      0.0063
## race_hisp_12m         0.0095      0.0095
## `factor(edu_12m)`1    0.0206      0.0206
## `factor(edu_12m)`2    0.0247      0.0247
## `factor(edu_12m)`3    0.0077      0.0077
## `factor(edu_12m)`4    0.0375      0.0375
##
## Summary of Balance for Matched Data:
##               Means Treated Means Control Std. Mean Diff. Var. Ratio
## distance              0.2868          0.2868          0.0001      1.0005
## age12                 41.1421         41.3067         -0.0137      0.9961
## female_list           0.6318          0.6291          0.0056          .
## english_list          0.9261          0.9349         -0.0337          .
## race_white_12m        0.8394          0.8519         -0.0340          .
## race_black_12m        0.0386          0.0346          0.0208          .
## race_hisp_12m         0.1098          0.0992          0.0338          .
## `factor(edu_12m)`1    0.1827          0.1803          0.0062          .
## `factor(edu_12m)`2    0.5143          0.5177         -0.0067          .
## `factor(edu_12m)`3    0.2158          0.2147          0.0027          .
## `factor(edu_12m)`4    0.0872          0.0873         -0.0006          .
##               eCDF Mean eCDF Max Std. Pair Dist.
## distance              0.0000      0.0010          0.0004
## age12                 0.0042      0.0096          0.0857
## female_list           0.0027      0.0027          0.0389
## english_list          0.0088      0.0088          0.1207
## race_white_12m        0.0125      0.0125          0.1117
## race_black_12m        0.0040      0.0040          0.0973
## race_hisp_12m         0.0106      0.0106          0.1292
## `factor(edu_12m)`1    0.0024      0.0024          0.0452
## `factor(edu_12m)`2    0.0034      0.0034          0.0458
## `factor(edu_12m)`3    0.0011      0.0011          0.0425
## `factor(edu_12m)`4    0.0002      0.0002          0.0256

```

```
##
## Sample Sizes:
##           Control Treated
## All       16063    6241
## Matched   6241     6241
## Unmatched 9822      0
## Discarded 0        0

# (c) balance diagnostics
# (c.1) Love plot: standardized mean differences, before vs after matching.
# stars = "std" Labels which differences are standardized vs raw - this silences
# cobalt's (innocuous) mixed continuous/binary warning.
love.plot(m_out1, stats = "mean.diffs", threshold = 0.1, abs = TRUE,
          var.order = "unadjusted", stars = "std",
          title = "Covariate Balance Before/After NN Matching (Medicaid)")
```

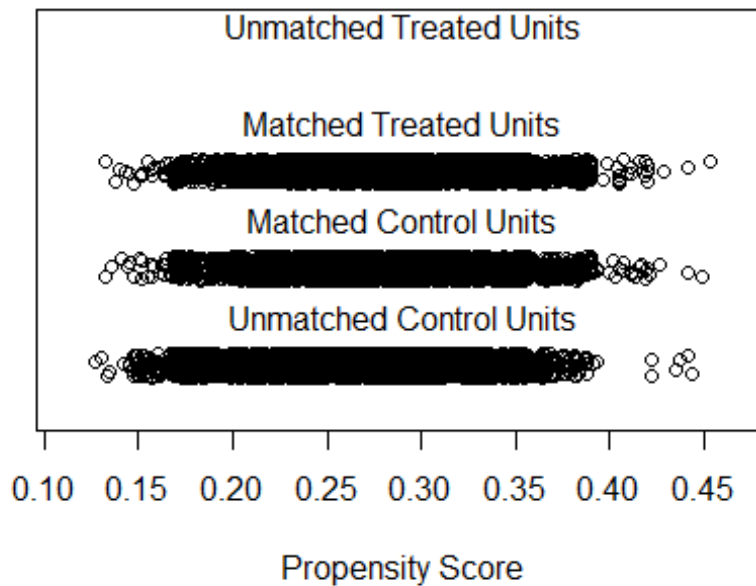
Covariate Balance Before/After NN Matching (Medicaid)



How to read it: each covariate has two dots. The "Unadjusted" dots (raw data) sit out to the right; after matching the "Adjusted" dots collapse inside the 0.10 rule-of-thumb band. So on observables the matched sample is now well balanced.

(c.2) Jitter plot: where matched/unmatched units sit on the PS support.
`plot(m_out1, type = "jitter", interactive = FALSE)`

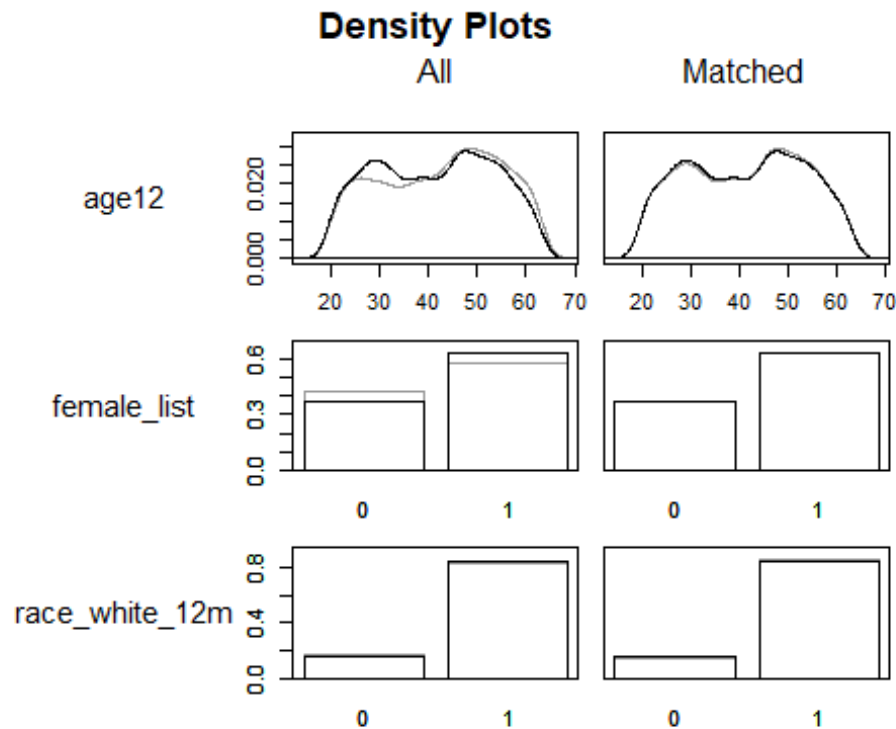
Distribution of Propensity Scores



*# How to read it: matched treated and matched control units cover the same range
of propensity scores (good common support). The unmatched controls are the
ones with no nearby treated unit, so they are dropped.*

*# (c.3) Density plot: the actual covariate distributions.
It makes balance tangible. Instead of one number per covariate, we see the
whole treated vs control distribution line up after matching.*

```
plot(m_out1, type = "density", interactive = FALSE,  
     which.xs = ~ age12 + female_list + race_white_12m)
```



How to read it: before matching the treated (solid) and control (dashed) curves can sit apart (imbalance). After matching they overlap almost on top of each other. Same message as the Love plot, but on the full distribution rather than just the mean.

(d) ATT after matching

```
m.data <- match_data(m_out1)
att_h <- lm(good_health ~ insured, data = m.data, weights = weights)
att_d <- lm(cost_any_owe_12m ~ insured, data = m.data, weights = weights)
print(clx(att_h, m.data$household_id)["insured", , drop = FALSE])
```

```
##           Estimate Std. Error  t value Pr(>|t|)
## insured -0.01009454 0.008980472 -1.124054 0.2610117
```

```
print(clx(att_d, m.data$household_id)["insured", , drop = FALSE])
```

```
##           Estimate Std. Error  t value    Pr(>|t|)
## insured -0.06617529 0.009069863 -7.296173 3.139724e-13
```

Reading: after all that balancing work, the matched estimate is essentially the same as the naive OLS - still ~ 0 for good_health. This is the key lesson: matching fixes the observables (we just saw the Love/density plots prove it), but enrollment is selected on unobservables (health, motivation, expected need) that no observational method can see. Good balance on X is not the same as identification.

---- (3) ITT (effect of the offer) -----

Now the clean estimate. We compare lottery winners vs losers, assignment was

```

# random. The factor(wave_survey12m) FE only sharpen precision, but the
# factor(numhh_list) FE are required for identification: the lottery gave larger
# households a higher win probability, so randomization holds only CONDITIONAL on
# household size. The estimand here is the effect of the offer of coverage, not of
# enrollment.

# Model: outcome = b0 + b1*treatment + (wave x hh-size FE) + u
# factor(wave_survey12m)*factor(numhh_list) = survey-wave x household-size FE
# weights = weight_12m (survey design / non-response weights).
itt_h <- lm(good_health ~ treatment + factor(wave_survey12m)*factor(numhh_list),
            data = da, weights = weight_12m)
itt_d <- lm(cost_any_owe_12m ~ treatment + factor(wave_survey12m)*factor(numhh_list),
            data = da, weights = weight_12m)
print(clx(itt_h, da$household_id)["treatment", , drop = FALSE])

##           Estimate Std. Error t value Pr(>|t|)
## treatment 0.0386944 0.007634702 5.068227 4.04603e-07

print(clx(itt_d, da$household_id)["treatment", , drop = FALSE])

##           Estimate Std. Error t value Pr(>|t|)
## treatment -0.05232376 0.007626034 -6.861202 6.998434e-12

## ---- (4) LATE / 2SLS -----
# Source: Imbens & Angrist (1994), Econometrica 62(2) [LATE]; Angrist & Pischke
# (2009), "Mostly Harmless Econometrics", ch.4; ivreg() from the AER
# package (Kleibergen & Zeileis 2008).
# The Lottery (treatment) instruments actual enrollment (insured). For a single
# instrument the 2SLS estimate equals the ITT divided by the first stage, i.e.
# the effect of enrollment for the compliers (those who enroll because they won).
if (requireNamespace("AER", quietly = TRUE)) {
  library(AER)
  late_h <- ivreg(good_health ~ insured + factor(wave_survey12m)*factor(numhh_list)
  |
  treatment + factor(wave_survey12m)*factor(numhh_list),
  data = da, weights = weight_12m)
  print(clx(late_h, da$household_id)["insured", , drop = FALSE])
}

## Loading required package: survival

##           Estimate Std. Error t value Pr(>|t|)
## insured 0.1328789 0.02642659 5.028227 4.987001e-07

# Reading: observational methods (OLS/Matching) understate the effect on
# self-reported health and understate the effect on debt; the clean experimental
# estimates are larger. Two faces of the same lesson: observational
# identification is unreliable here.

# ---- Figure: observational vs experimental -----

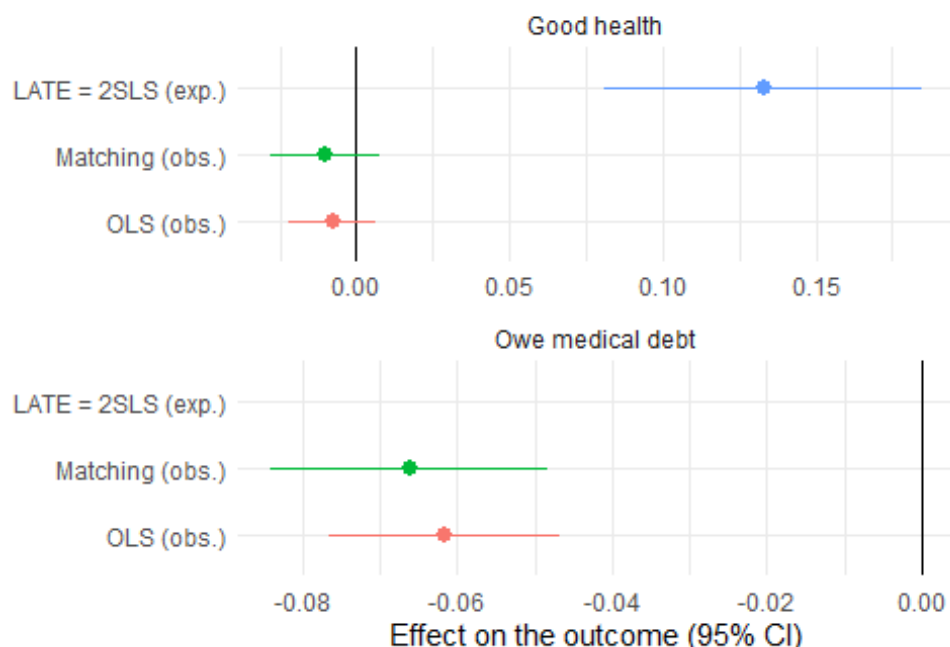
```

```

# Everything above, on one axis. For the two headline outcomes we line up the
# estimators that all target the same estimand, the effect of enrollment, and
# let the reader see the selection bias directly.
# Important: we deliberately do not put the ITT on this axis. The ITT is the
# effect of the offer (a different estimand).
# LATE was only estimated for good_health above, so it appears only in
# that panel.
mk <- function(out, meth, model, clus) {
  ct <- clx(model, clus)["insured", ]
  data.frame(Outcome = out, Method = meth,
             est = ct["Estimate"], se = ct["Std. Error"], row.names = NULL)
}
cmp <- rbind(
  mk("Good health", "OLS (obs.)", ols_h, da$household_id),
  mk("Good health", "Matching (obs.)", att_h, m.data$household_id),
  mk("Owe medical debt", "OLS (obs.)", ols_d, da$household_id),
  mk("Owe medical debt", "Matching (obs.)", att_d, m.data$household_id))
if (exists("late_h")) {
  cmp <- rbind(cmp, mk("Good health", "LATE = 2SLS (exp.)", late_h, da$household_id))
}
cmp$ci_l <- cmp$est - 1.96 * cmp$se
cmp$ci_h <- cmp$est + 1.96 * cmp$se
cmp$Method <- factor(cmp$Method,
                     levels = c("OLS (obs.)", "Matching (obs.)", "LATE = 2SLS (exp.)"))
ggplot(cmp, aes(x = est, y = Method, colour = Method)) +
  geom_vline(xintercept = 0) +
  geom_pointrange(aes(xmin = ci_l, xmax = ci_h), show.legend = FALSE) +
  facet_wrap(~ Outcome, scales = "free_x", ncol = 1) +
  labs(title = "Effect of Medicaid ENROLLMENT: observational vs experimental",
       subtitle = "OLS & Matching (blind to unobservables) vs LATE (lottery-identified). ITT omitted on purpose: different estimand.",
       x = "Effect on the outcome (95% CI)", y = NULL) +
  theme_minimal()

```

Effect of Medicaid ENROLLMENT: observatic OLS & Matching (blind to unobservables) vs LATE (lotter



How to read it: focus on the good_health panel, where all three estimators are
present. OLS and Matching agree with each other (both ~ 0) but sit faar from the
experimental LATE ($\sim +0.13$), and the gap is much wider than the confidence
intervals. So it is selection bias, not noise. In the debt panel we only have
the two observational estimates (no late was run for debt). They agree with
each other, which is reassuring but says nothing about unobserved selection.
Bottom Line: adjusting for observables does not recover the
experimental answer, because enrollment is selected on unobservables.

III - MAIN RESULT: ITT across the three outcome domains

We estimate the ITT for every outcome in three QJE families and summarise each
family with one standardized index (so that many outcomes do not inflate the
number of tests). Each ITT is a single `lm()` with clustered SE, as above.

```
fam <- list(
  "Health care use" = c("rx_num_mod_12m", "doc_num_mod_12m", "er_num_mod_12m", "hosp
_num_mod_12m"),
  "Financial strain" = c("cost_any_oop_12m", "cost_any_owe_12m", "cost_borrow_12m", "
cost_refused_12m"),
  "Health" = c("good_health", "health_notpoor", "health_improved",
               "notbaddays_tot", "notbaddays_phys", "notbaddays_ment", "nod
ep_screen"))

# --- ITT per outcome -----
itt_one <- function(y) {
```



```

m <- lm(as.formula(paste(y, "~ treatment + factor(wave_survey12m)*factor(numhh_
list)")),
      data = da, weights = weight_12m)
ct <- clx(m, da$household_id["treatment", ])
cm <- weighted.mean(da[[y]][da$treatment == 0], da$weight_12m[da$treatment == 0],
na.rm = TRUE)
data.frame(outcome = y, control_mean = cm,
      est = ct["Estimate"], se = ct["Std. Error"], p = ct["Pr(>|t|)"])
}
res <- data.frame()
for (k in names(fam)) {
  block <- do.call(rbind, lapply(fam[[k]], itt_one))
  block$domain <- k
  res <- rbind(res, block)
}
res$ci_l <- res$est - 1.96 * res$se
res$ci_h <- res$est + 1.96 * res$se
cat("\n--- ITT by outcome (control mean, estimate, SE, p) ---\n")

##
## --- ITT by outcome (control mean, estimate, SE, p) ---

print(round(res[, c("est", "se", "p", "control_mean")], 4))

##
##      est      se      p control_mean
## Estimate  0.0999 0.0508 0.0491      2.3183
## Estimate1  0.3141 0.0536 0.0000      1.9142
## Estimate2  0.0074 0.0161 0.6452      0.4697
## Estimate3  0.0062 0.0062 0.3112      0.0975
## Estimate4 -0.0580 0.0077 0.0000      0.5545
## Estimate11 -0.0523 0.0076 0.0000      0.5966
## Estimate21 -0.0447 0.0073 0.0000      0.3641
## Estimate31 -0.0105 0.0041 0.0102      0.0813
## Estimate7  0.0387 0.0076 0.0000      0.5482
## Estimate12 0.0288 0.0051 0.0000      0.8597
## Estimate22 0.0328 0.0067 0.0000      0.7137
## Estimate32 0.3810 0.1618 0.0185     21.8623
## Estimate41 0.4592 0.1746 0.0085     20.3289
## Estimate5  0.6029 0.1843 0.0011     18.7381
## Estimate6  0.0227 0.0071 0.0014      0.6710

# --- Standardized treatment-effect index per domain -----
# index = mean of the z-scored components, standardized on the control group.
# This is the QJE multiple-testing summary (Kling, Liebman & Katz 2007, ECMA):
# one test per domain instead of many. (Standardizing/weighted moments are
# base-R. The index construction itself follows the QJE code.)
make_index <- function(vars, data) {
  ctrl <- subset(data, treatment == 0)
  Z <- sapply(vars, function(v) {
    mu <- weighted.mean(ctrl[[v]], ctrl$weight_12m, na.rm = TRUE)
    sdv <- sqrt(weighted.mean((ctrl[[v]] - mu)^2, ctrl$weight_12m, na.rm = TRUE))
    (data[[v]] - mu) / sdv
  })

```

```

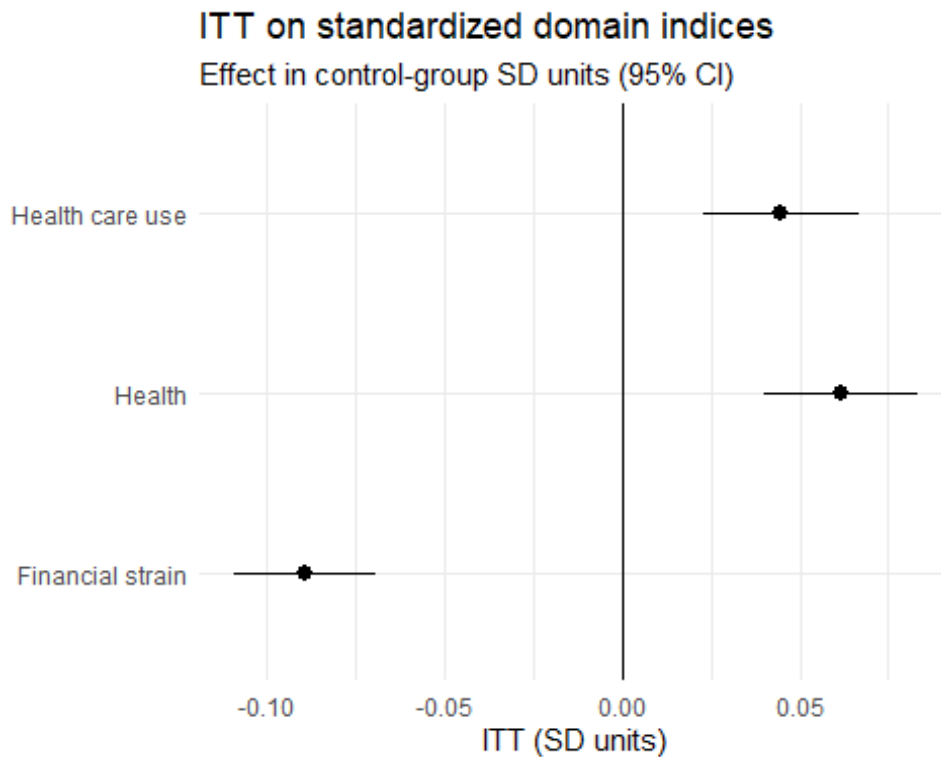
}))
rowMeans(Z, na.rm = TRUE)
}
da$idx_use <- make_index(fam[["Health care use"]], da)
da$idx_fin <- make_index(fam[["Financial strain"]], da)
da$idx_health <- make_index(fam[["Health"]], da)
# Orientation:
# the health items are oriented in the data-prep script (prep_data_oregon.R) so th
at
# "higher = better". The financial-strain items were deliberately not flipped.
# They keep their natural direction, so higher = more strain (more out-of-pocket,
# more debt, more borrowing, refused care). Therefore a negative ITT on idx_fin is
# a good outcome: the offer reduces financial strain. Do not misread the minus sig
n as "harm".

idx_tab <- data.frame()
for (ix in c("idx_use", "idx_fin", "idx_health")) {
  m <- lm(as.formula(paste(ix, "~ treatment + factor(wave_survey12m)*factor(numhh
_list)")),
        data = da, weights = weight_12m)
  ct <- clx(m, da$household_id["treatment", ])
  idx_tab <- rbind(idx_tab, data.frame(index = ix,
        est = ct["Estimate"], se = ct["Std. Error"], p = ct["Pr(>|t|)
"]))
}
idx_tab$ci_l <- idx_tab$est - 1.96 * idx_tab$se
idx_tab$ci_h <- idx_tab$est + 1.96 * idx_tab$se
print(round(idx_tab[, c("est", "se", "p")], 4))

##           est      se      p
## Estimate   0.0446 0.0112 1e-04
## Estimate1 -0.0892 0.0102 0e+00
## Estimate2  0.0618 0.0111 0e+00

# Figure: domain-index effects with 95% CI (comparable across domains).
idx_tab$label <- c("Health care use", "Financial strain", "Health")
ggplot(idx_tab, aes(x = est, y = label)) +
  geom_vline(xintercept = 0) +
  geom_pointrange(aes(xmin = ci_l, xmax = ci_h)) +
  labs(title = "ITT on standardized domain indices",
        subtitle = "Effect in control-group SD units (95% CI)",
        x = "ITT (SD units)", y = NULL) +
  theme_minimal()

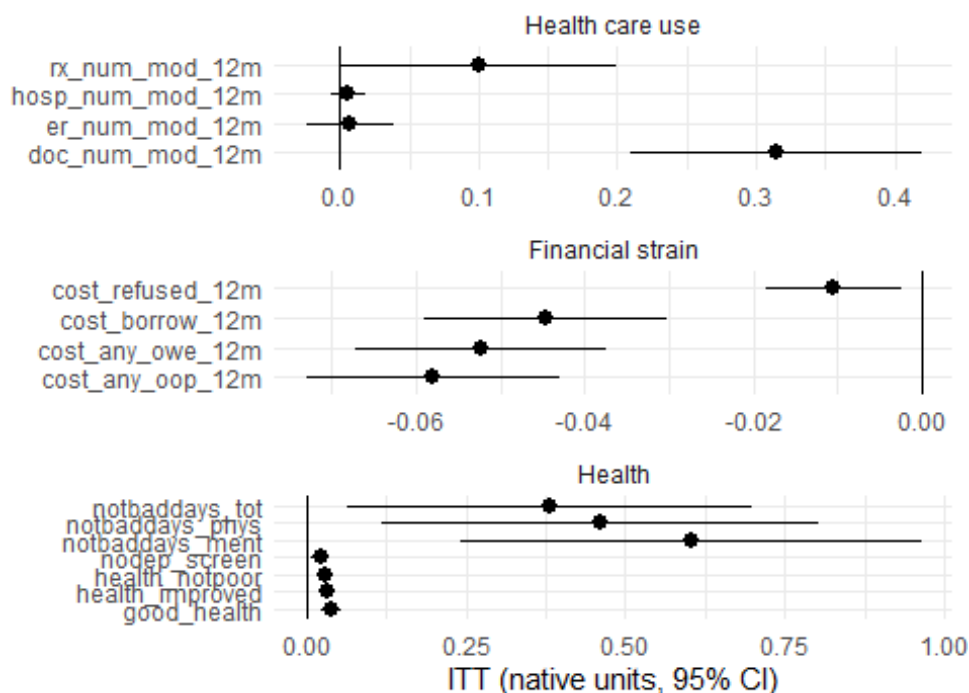
```



How to read it: each dot is the ITT on one domain index, the bar is its 95% CI,
 # the vertical line is the zero (no-effect) reference. Units are control-group
 # SDs, so the three effects are directly comparable in size.
 # - Health care use -> dot & CI to the RIGHT of 0 => the offer raises use
 # - Health -> dot & CI to the RIGHT of 0 => self-reported health up
 # - Financial strain -> dot & CI to the LEFT of 0 => less strain
 # A CI that does not cross the zero line is what "significant at 5%" looks like.

```
# Figure: per-component ITT, faceted by domain
res$domain <- factor(res$domain, levels = names(fam))
ggplot(res, aes(x = est, y = outcome)) +
  geom_vline(xintercept = 0) +
  geom_pointrange(aes(xmin = ci_l, xmax = ci_h)) +
  facet_wrap(~ domain, scales = "free", ncol = 1) +
  labs(title = "Intention-to-treat effects by outcome and domain",
       x = "ITT (native units, 95% CI)", y = NULL) +
  theme_minimal()
```

Intention-to-treat effects by outcome and dc



How to read it: same coefficient-plot logic, now one row per raw outcome,
 # faceted by domain (native units, free x-scales so each domain is legible).
 # Reading: care use is up, driven by doctor visits and prescriptions; financial
 # strain is down across the board (the most robust result); health is up, but
 # the health gain is concentrated in mental health and self-reported measures.
 # Be honest about the limit here: the OHIE in-person exam found no significant
 # change in objective physical biomarkers (blood pressure, cholesterol, HbA1c)
 # (Baicker et al. 2013, NEJM 368:1713-1722). So "health improved" means mainly
 # perceived and mental health, not measured physical disease.

 # IV - HETEROGENEITY BY AGE (CATE)
 #####
 # Is the effect stronger for some groups, e.g. the older (50-64), who have more
 # chronic conditions? We compare the ITT in two subgroups and test the
 # difference formally with an interaction term treatment x older.

```
cate_one <- function(ix) {
  # subgroup ITTs
  m_y <- lm(as.formula(paste(ix, "~ treatment + factor(wave_survey12m)*factor(numh
h_list)")),
            data = subset(da, younger == 1), weights = weight_12m)
  m_o <- lm(as.formula(paste(ix, "~ treatment + factor(wave_survey12m)*factor(numh
h_list)")),
            data = subset(da, older == 1), weights = weight_12m)
  cy <- clx(m_y, subset(da, younger == 1)$household_id["treatment", ])
  co <- clx(m_o, subset(da, older == 1)$household_id["treatment", ])
  # formal heterogeneity test: coefficient on treatment:older
}
```

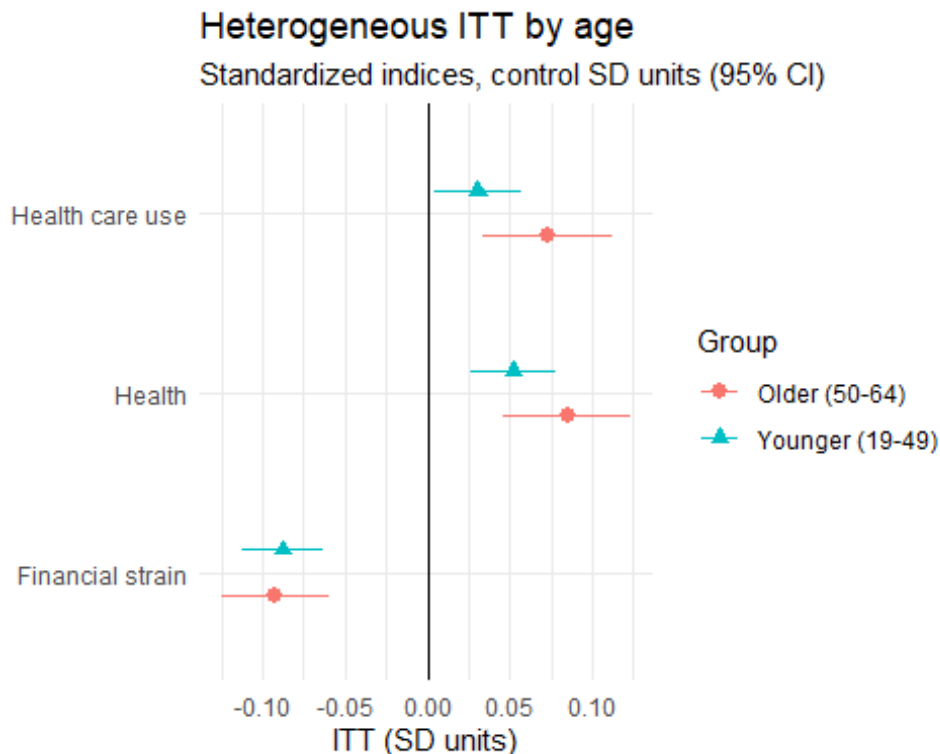
```

m_i <- lm(as.formula(paste(ix, "~ treatment*older + factor(wave_survey12m)*factor(numhh_list)")),
          data = da, weights = weight_12m)
ci <- clx(m_i, da$household_id)["treatment:older", ]
data.frame(index = ix,
            itt_young = cy["Estimate"], itt_old = co["Estimate"],
            diff = ci["Estimate"], se_diff = ci["Std. Error"], p_het = ci["Pr(>|t
|)"])
}
cate <- do.call(rbind, lapply(c("idx_use", "idx_fin", "idx_health"), cate_one))
print(round(cate[, c("itt_young", "itt_old", "diff", "p_het")], 4))

##           itt_young itt_old    diff p_het
## Estimate      0.0306  0.0731  0.0421 0.0697
## Estimate1    -0.0878 -0.0924 -0.0008 0.9682
## Estimate2      0.0524  0.0852  0.0334 0.1426

# Figure: subgroup ITT by age (dodged point-ranges).
cate$label <- c("Health care use", "Financial strain", "Health")
# helper: clustered SE of the subgroup ITT, recomputed for the error bars
get_se <- function(ix, keep) {
  sub <- subset(da, keep)
  m <- lm(as.formula(paste(ix, "~ treatment + factor(wave_survey12m)*factor(numhh_
list)")),
          data = sub, weights = weight_12m)
  clx(m, sub$household_id)["treatment", "Std. Error"]
}
indices <- c("idx_use", "idx_fin", "idx_health")
plot_df <- rbind(
  data.frame(label = cate$label, Group = "Younger (19-49)", est = cate$itt_young,
             se = sapply(indices, function(ix) get_se(ix, da$younger == 1))),
  data.frame(label = cate$label, Group = "Older (50-64)", est = cate$itt_old,
             se = sapply(indices, function(ix) get_se(ix, da$older == 1))))
plot_df$ci_l <- plot_df$est - 1.96 * plot_df$se
plot_df$ci_h <- plot_df$est + 1.96 * plot_df$se
ggplot(plot_df, aes(x = est, y = label, colour = Group, shape = Group)) +
  geom_vline(xintercept = 0) +
  geom_pointrange(aes(xmin = ci_l, xmax = ci_h),
                 position = position_dodge(width = 0.5)) +
  labs(title = "Heterogeneous ITT by age",
       subtitle = "Standardized indices, control SD units (95% CI)",
       x = "ITT (SD units)", y = NULL) +
  theme_minimal()

```



*# Reading: point estimates suggest larger effects for the older;
 # financial protection is essentially age-invariant.
 # Caution: subgroups split the sample -> low power -> a non-significant
 # interaction is not proof of "no heterogeneity".*

*#####
 # V - RANDOMIZATION INFERENCE (robustness of the inference)
 #####
 # Instead of relying only on the (asymptotic) clustered SE, we rebuild the null
 # distribution from the design itself. We re-run the lottery thousands of times
 # on placebo, preserving how it was randomized (at the household level, within
 # household-size strata), and see how often a placebo effect is as large as ours.
 # This is finite-sample valid inference.
 # It is out robustness check, built on the Fisher randomization-test logic
 # in Imbens & Rubin (2015), ch.5.
 # Note this is not the same as what the QJE does for multiple
 # testing: that paper uses the Westfall-Young free step-down correction, which
 # serves a different purpose (controlling family-wise error across many
 # outcomes).*

```
ri_pvalue <- function(ix, data, n_perm = 2000, seed = 42) {
  set.seed(seed)
  d <- data[!is.na(data[[ix]]), ]
  # observed ITT (point estimate from the same weighted lm)
  obs <- coef(lm(as.formula(paste(ix, "~ treatment + factor(wave_survey12m)*factor(
numhh_list)")),
    data = d, weights = weight_12m))["treatment"]
  # household-level table with empirical treatment probability within hh-size
```

```

hh <- aggregate(treatment ~ household_id + numhh_list, data = d, FUN = function
(x) x[1])
pr <- tapply(hh$treatment, hh$numhh_list, mean)           # P(treat) per stratum
hh$p <- pr[as.character(hh$numhh_list)]
row_hh <- match(d$household_id, hh$household_id)         # map each row -> its
household
count <- 0
for (i in seq_len(n_perm)) {
  sim_hh <- as.numeric(runif(nrow(hh)) <= hh$p)           # placebo treatment pe
r household
  d$sim <- sim_hh[row_hh]                                 # propagate to all mem
bers
  b <- coef(lm(as.formula(paste(ix, "~ sim + factor(wave_survey12m)*factor(numhh
_list)")),
              data = d, weights = weight_12m))["sim"]
  if (!is.na(b) && abs(b) >= abs(obs)) count <- count + 1
}
c(observed = unname(obs), ri_p = (1 + count) / (1 + n_perm))
}
for (ix in c("idx_use", "idx_fin", "idx_health")) {
  out <- ri_pvalue(ix, da)
  cat(sprintf("%-11s observed ITT = %.4f    RI p-value = %.4f\n", ix, out["observe
d"], out["ri_p"]))
}

## idx_use      observed ITT = +0.0446    RI p-value = 0.0005
## idx_fin      observed ITT = -0.0892    RI p-value = 0.0005
## idx_health   observed ITT = +0.0618    RI p-value = 0.0005

# Note: this loop refits the model 2000x per index.
# Reading: the observed effects are more extreme than essentially all placebos
# (p ~ 0.0005), confirming the clustered-SE conclusions without leaning on the
# normal approximation.

#####
# VI - BRIDGE TO PART I: MORTALITY
#####
# Part I studied insurance -> mortality on state-level cross-sectional OLS and
# could not credibly identify an effect (OVB, reverse causality, selection on
# unobservables). Here we estimate the same outcome under randomization.
# Mortality (postn_death = died after lottery notification) is administrative and
# observed for everyone, so we use the full lottery sample with draw + household-
# size FE and clustered SE - NOT the weighted survey sample. This is lm + cluster
d
# SE only: no method beyond what is already used above (no biomarkers, no Lee boun
ds).
dt$alive <- 1 - dt$postn_death                          # 1 = alive at the end of the study windo
w
mort <- lm(alive ~ treatment + factor(draw_lottery) + factor(numhh_list), data = d
t)
print(clx(mort, dt$household_id)[ "treatment", , drop = FALSE])

```

```
##           Estimate Std. Error   t value Pr(>|t|)
## treatment 0.0003198884 0.000679277 0.4709248 0.6376958
```

*# Reading: ITT on survival = +0.0003 (clustered SE 0.0007, p = 0.64), a precise
null, not evidence of "no effect". Control-arm mortality is only ~0.85%, so the
study is underpowered for a rare event over a ~16-month horizon. The 95% CI on
survival [-0.0010, +0.0017], re-expressed against control mortality, spans
roughly a 19% mortality reduction to a 12% increase: it cannot rule out a
clinically meaningful mortality benefit. This mirrors Finkelstein et al. (2012),
who likewise could not detect a mortality effect, and it is the cleanest bridge
to Part I: the exact outcome that cross-sectional OLS could not pin down, now
estimated from a randomized experiment.*

#####

CONCLUSION

#####

The core forms one arc:

*# (I) the Lottery is a valid experiment -> first stage, balance, non-response
(II) observational methods fail -> OLS/Matching vs LATE
(III) the causal effects -> ITT + standardized domain indices
(IV) for whom -> CATE by age
(V) how sure we are -> randomization inference
(VI) shared-outcome bridge -> mortality ITT
#*

*# Faithful replication of the survey core of Finkelstein et al. (2012, QJE), plus
our own selection-bias demonstration (OLS/Matching vs LATE) as the conceptual
and empirical bridge to Part I.*

#####